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1. Dynamics of Feature Categorization

Daniel Martí, John Rinzel

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In visual and auditory scenes, we are able to identify shared features among sensory objects and group them according to their similarity. This grouping is preattentive and fast and is thought of as an elementary form of categorization by which objects sharing similar features are clustered in some abstract perceptual space. It is unclear what neuronal mechanisms underlie this fast categorization. Here we propose a neuromechanistic model of fast feature categorization based on the framework of continuous attractor networks. The mechanism for category formation does not rely on learning and is based on biologically plausible assumptions, for example, the existence of populations of neurons tuned to feature values, feature-specific interactions, and subthreshold-evoked responses upon the presentation of single objects. When the network is presented with a sequence of stimuli characterized by some feature, the network sums the evoked responses and provides a running estimate of the distribution of features in the input stream. If the distribution of features is structured into different components or peaks (i.e., is multimodal), recurrent excitation amplifies the response of activated neurons, and categories are singled out as emerging localized patterns of elevated neuronal activity (bumps), centered at the centroid of each cluster. The emergence of bump states through sequential, subthreshold activation and the dependence on input statistics is a novel application of attractor networks. We show that the extraction and representation of multiple categories are facilitated by the rich attractor structure of the network, which can sustain multiple stable activity patterns for a robust range of connectivity parameters compatible with cortical physiology.

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2. Stochastic Hodgkin-Huxley Equations with Colored Noise Terms in the Conductances

Marifi Güler

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The excitability of cells is facilitated by voltage-gated ion channels. These channels accommodate a multiple number of gates individually. The possible impact of that gate multiplicity on the cell's function, specifically when the membrane area is of limited size, was investigated in the author's prior work (Güler, 2011). There, it was found that a nontrivially persistent correlation takes place between the transmembrane voltage fluctuations (also between the fluctuations in the gating variables) and the component of open channel fluctuations attributed to the gate multiplicity. This nontrivial phenomenon was found to be playing a major augmentative role for the elevation of excitability and spontaneous firing in small cells. In addition, the same phenomenon was found to be enhancing spike coherence significantly. Here we extend Fox and Lu's (1994) stochastic Hodgkin-Huxley equations by incorporating colored noise terms into the conductances there to obtain a formalism capable of capturing the addressed cross-correlations. Statistics of spike generation, spike coherence, firing efficiency, latency, and jitter from the articulated set of equations are found to be highly accurate in comparison with the corresponding statistics from the exact microscopic Markov simulations. This way, it is demonstrated vividly that our formulation overcomes the inherent inadequacy of

the Fox and Lu equations. Finally, a recently proposed diffusion approximation method (Linaro, Storace, & Giugliano, 2011) is taken into consideration, and a discussion on its character is pursued.

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3. Statistical Computer Model Analysis of the Reciprocal and Recurrent Inhibitions of the Ia-EPSP in α-Motoneurons

Gérard Gradwohl, Yoram Grossman

https://doi.org/10.1162/neco_a_00375 | pp. 75-100

We simulate the inhibition of Ia-glutamatergic excitatory postsynaptic potential (EPSP) by preceding it with glycinergic recurrent (REN) and reciprocal (REC) inhibitory postsynaptic potentials (IPSPs). The inhibition is evaluated in the presence of voltage-dependent conductances of sodium, delayed rectifier potassium, and slow potassium in five α-motoneurons (MNs). We distribute the channels along the neuronal dendrites using, alternatively, a density function of exponential rise (ER), exponential decay (ED), or a step function (ST). We examine the change in EPSP amplitude, the rate of rise (RR), and the time integral (TI) due to inhibition. The results yield six major conclusions. First, the EPSP peak and the kinetics depending on the time interval are either amplified or depressed by the REC and REN shunting inhibitions. Second, the mean EPSP peak, its TI, and RR inhibition of ST, ER, and ED distributions turn out to be similar for analogous ranges of G. Third, for identical G, the large variations in the parameters' values can be attributed to the sodium conductance step (g(Na_step)) and the active dendritic area. We find that small g(Na_step) on a few dendrites maintains the EPSP peak, its TI, and RR inhibition similar to the passive state, but high g(Na_step) on many dendrites decrease the inhibition and sometimes generates even an excitatory effect. Fourth, the MN's input resistance does not alter the efficacy of EPSP inhibition. Fifth, the REC and REN inhibitions slightly change the EPSP peak and its RR. However, EPSP TI is depressed by the REN inhibition more than the REC inhibition. Finally, only an inhibitory effect shows up during the EPSP TI inhibition, while there are both inhibitory and excitatory impacts on the EPSP peak and its RR.

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4. Point-Process Principal Components Analysis via Geometric Optimization

Victor Solo, Syed Ahmed Pasha

https://doi.org/10.1162/neco_a_00382 | pp. 101-122

There has been a fast-growing demand for analysis tools for multivariate point-process data driven by work in neural coding and, more recently, high-frequency finance. Here we develop a true or exact (as opposed to one based on time binning) principal components analysis for preliminary processing of multivariate point processes. We provide a maximum likelihood estimator, an algorithm for maximization involving steepest ascent on two Stiefel manifolds, and novel constrained asymptotic analysis. The method is illustrated with a simulation and compared with a binning approach.

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5. A Spiking Neural Model for Stable Reinforcement of Synapses Based on Multiple Distal Rewards

Michael J. O'Brien, Narayan Srinivasa

https://doi.org/10.1162/neco_a_00387 | pp. 123-156

In this letter, a novel critic-like algorithm was developed to extend the synaptic plasticity rule described in Florian (2007) and Izhikevich (2007) in order to solve the problem of learning multiple distal rewards simultaneously. The system is augmented with short-term plasticity (STP) to stabilize the learning dynamics, thereby increasing the system's learning capacity. A theoretical threshold is estimated for the number of distal rewards that this system can learn. The validity of the novel algorithm was verified by computer simulations.

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6. A Model of the Differential Representation of Signal Novelty in the Local Field Potentials and Spiking Activity of the Ventrolateral Prefrontal Cortex

Jung H. Lee, Joji Tsunada, Yale E. Cohen

https://doi.org/10.1162/neco_a_00388 | pp. 157-185

Local field potentials (LFPs) and spiking activity reflect different types of information procssing. For example, neurophysiological studies indicate that signal novelty in the ventrolateral prefrontal cortex is differentially represented by LFPs and spiking activity: LFPs habituate to repeated stimulus presentations, whereas spiking activity does not. The neural mechanisms that allow for this differential representation between LFPs and spiking activity are not clear. Here, we model and simulate LFPs and spiking activity of neurons in the ventrolateral prefrontal cortex in order to elucidate potential mechanisms underlying this differential representation. We demonstrate that dynamic negative-feedback loops cause LFPs to habituate in response to repeated presentations of the same stimulus while spiking activity is maintained. This disassociation between LFPs and spiking activity may be a mechanism by which LFPs code stimulus novelty, whereas spiking activity carries abstract information, such as category membership and decision-related activity.

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7. Computing Sparse Representations of Multidimensional Signals Using Kronecker Bases

César F. Caiafa, Andrzej Cichocki

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Recently there has been great interest in sparse representations of signals under the assumption that signals (data sets) can be well approximated by a linear combination of few elements of a known basis (dictionary). Many algorithms have been developed to find such representations for one-dimensional signals (vectors), which requires finding the sparsest solution of an underdetermined linear system of algebraic equations. In this letter, we generalize the theory of sparse representations of vectors to multiway arrays (tensors)--signals with a multidimensional structure--by using the Tucker model. Thus, the problem is reduced to solving a large-scale underdetermined linear system of equations possessing a Kronecker structure, for which we have developed a greedy algorithm, Kronecker-OMP, as a generalization of the classical orthogonal matching pursuit (OMP) algorithm for vectors. We also introduce the concept of multiway block-sparse representation of N-way arrays and develop a new greedy algorithm that exploits not only the Kronecker structure but also block sparsity. This allows us to derive a very fast and memory-efficient algorithm called N-BOMP (N-way block OMP). We theoretically demonstrate that under the block-sparsity assumption, our N-BOMP algorithm not only has a considerably lower complexity but is also more precise than the classic OMP algorithm. Moreover, our algorithms can be used for very large-scale problems, which are intractable using standard approaches. We provide several simulations illustrating our

results and comparing our algorithms to classical algorithms such as OMP and BP (basis pursuit) algorithms. We also apply the N-BOMP algorithm as a fast solution for the compressed sensing (CS) problem with large-scale data sets, in particular, for 2D compressive imaging (CI) and 3D hyperspectral CI, and we show examples with real-world multidimensional signals.

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8. ANUBIS: Artificial Neuromodulation Using a Bayesian Inference System

Benjamin J. Smith, Chakravarthini M. Saaj, Elie Allouis

https://doi.org/10.1162/neco_a_00376 | pp. 221-258

Gain tuning is a crucial part of controller design and depends not only on an accurate understanding of the system in question, but also on the designer's ability to predict what disturbances and other perturbations the system will encounter throughout its operation. This letter presents ANUBIS (artificial neuromodulation using a Bayesian inference system), a novel biologically inspired technique for automatically tuning controller parameters in real time. ANUBIS is based on the Bayesian brain concept and modifies it by incorporating a model of the neuromodulatory system comprising four artificial neuromodulators. It has been applied to the controller of EchinoBot, a prototype walking rover for Martian exploration. ANUBIS has been implemented at three levels of the controller; gait generation, foot trajectory planning using Bézier curves, and foot trajectory tracking using a terminal sliding mode controller. We compare the results to a similar system that has been tuned using a multilayer perceptron. The use of Bayesian inference means that the system retains mathematical interpretability, unlike other intelligent tuning techniques, which use neural networks, fuzzy logic, or evolutionary algorithms. The simulation results show that ANUBIS provides significant improvements in efficiency and adaptability of the three controller components; it allows the robot to react to obstacles and uncertainties faster than the system tuned with the MLP, while maintaining stability and accuracy. As well as advancing rover autonomy, ANUBIS could also be applied to other situations where operating conditions are likely to change or cannot be accurately modeled in advance, such as process control. In addition, it demonstrates one way in which neuromodulation could fit into the Bayesian brain framework.

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9. Multilayer Perceptron Classification of Unknown Volatile Chemicals from the Firing Rates of Insect Olfactory Sensory Neurons and Its Application to Biosensor Design

Luqman R. Bachtiar, Charles P. Unsworth, Richard D. Newcomb, Edmund J. Crampin

https://doi.org/10.1162/neco_a_00386 | pp. 259-287

In this letter, we use the firing rates from an array of olfactory sensory neurons (OSNs) of the fruit fly, Drosophila melanogaster, to train an artificial neural network (ANN) to distinguish different chemical classes of volatile odorants. Bootstrapping is implemented for the optimized networks, providing an accurate estimate of a network's predicted values. Initially a simple linear predictor was used to assess the complexity of the data and was found to provide low prediction performance. A nonlinear ANN in the form of a single multilayer perceptron (MLP) was also used, providing a significant increase in prediction performance. The effect of the number of hidden layers and hidden neurons of the MLP was investigated and found to be effective in enhancing network performance with both a single and a double hidden layer investigated separately. A hybrid array of MLPs was investigated and compared against the single MLP architecture. The

hybrid MLPs were found to classify all vectors of the validation set, presenting the highest degree of prediction accuracy. Adjustment of the number of hidden neurons was investigated, providing further performance gain. In addition, noise injection was investigated, proving successful for certain network designs. It was found that the best-performing MLP was that of the double-hidden-layer hybrid MLP network without the use of noise injection. Furthermore, the level of performance was examined when different numbers of OSNs used were varied from the maximum of 24 to only 5 OSNs. Finally, the ideal OSNs were identified that optimized network performance. The results obtained from this study provide strong evidence of the usefulness of ANNs in the field of olfaction for the future realization of a signal processing back end for an artificial olfactory biosensor.

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10. Impact of Correlated Neural Activity on Decision-Making Performance

Nicholas Cain, Eric Shea■Brown

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Stimulus from the environment that guides behavior and informs decisions is encoded in the firing rates of neural populations. Neurons in the populations, however, do not spike independently: spike events are correlated from cell to cell. To what degree does this apparent redundancy have an impact on the accuracy with which decisions can be made and the computations required to optimally decide? We explore these questions for two illustrative models of correlation among cells. Each model is statistically identical at the level of pairwise correlations but differs in higher-order statistics that describe the simultaneous activity of larger cell groups. We find that the presence of correlations can diminish the performance attained by an ideal decision maker to either a small or large extent, depending on the nature of the higher-order correlations. Moreover, although this optimal performance can in some cases be obtained using the standard integration-to-bound operation, in others it requires a nonlinear computation on incoming spikes. Overall, we conclude that a given level of pairwise correlations, even when restricted to identical neural populations, may not always indicate redundancies that diminish decision-making performance.

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11. Dynamical Movement Primitives: Learning Attractor Models for Motor Behaviors

Auke Jan Ijspeert, Jun Nakanishi, H. Hoffmann, Peter Pástor, Stefan Schaal

https://doi.org/10.1162/neco_a_00393 | pp. 328-373

Nonlinear dynamical systems have been used in many disciplines to model complex behaviors, including biological motor control, robotics, perception, economics, traffic prediction, and neuroscience. While often the unexpected emergent behavior of nonlinear systems is the focus of investigations, it is of equal importance to create goal-directed behavior (e.g., stable locomotion from a system of coupled oscillators under perceptual guidance). Modeling goal-directed behavior with nonlinear systems is, however, rather difficult due to the parameter sensitivity of these systems, their complex phase transitions in response to subtle parameter changes, and the difficulty of analyzing and predicting their long-term behavior; intuition and time-consuming parameter tuning play a major role. This letter presents and reviews dynamical movement primitives, a line of research for modeling attractor behaviors of autonomous nonlinear dynamical systems with the help of statistical learning techniques. The essence of our approach is to start with a simple dynamical system, such as a set of linear differential equations, and transform those into a weakly nonlinear

system with prescribed attractor dynamics by means of a learnable autonomous forcing term. Both point attractors and limit cycle attractors of almost arbitrary complexity can be generated. We explain the design principle of our approach and evaluate its properties in several example applications in motor control and robotics.

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12. Stochastic Optimal Control as a Theory of Brain-Machine Interface Operation

Manuel Lagang, Lakshminarayan Srinivasan

https://doi.org/10.1162/neco_a_00394 | pp. 374-417

The closed-loop operation of brain-machine interfaces (BMI) provides a framework to study the mechanisms behind neural control through a restricted output channel, with emerging clinical applications to stroke, degenerative disease, and trauma. Despite significant empirically driven improvements in closed-loop BMI systems, a fundamental, experimentally validated theory of closed-loop BMI operation is lacking. Here we propose a compact model based on stochastic optimal control to describe the brain in skillfully operating canonical decoding algorithms. The model produces goal-directed BMI movements with sensory feedback and intrinsically noisy neural output signals. Various experimentally validated phenomena emerge naturally from this model, including performance deterioration with bin width, compensation of biased decoders, and shifts in tuning curves between arm control and BMI control. Analysis of the model provides insight into possible mechanisms underlying these behaviors, with testable predictions. Spike binning may erode performance in part from intrinsic control-dependent constraints, regardless of decoding accuracy. In compensating decoder bias, the brain may incur an energetic cost associated with action potential production. Tuning curve shifts, seen after the mastery of a BMI-based skill, may reflect the brain's implementation of a new closed-loop control policy. The direction and magnitude of tuning curve shifts may be altered by decoder structure, ensemble size, and the costs of closed-loop control. Looking forward, the model provides a framework for the design and simulated testing of an emerging class of BMI algorithms that seek to directly exploit the presence of a human in the loop.

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13. Accelerated Spike Resampling for Accurate Multiple Testing Controls

Matthew Tom Harrison

https://doi.org/10.1162/neco_a_00399 | pp. 418-449

Controlling for multiple hypothesis tests using standard spike resampling techniques often requires prohibitive amounts of computation. Importance sampling techniques can be used to accelerate the computation. The general theory is presented, along with specific examples for testing differences across conditions using permutation tests and for testing pairwise synchrony and precise lagged-correlation between many simultaneously recorded spike trains using interval jitter.

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14. A Spike-Timing-Based Integrated Model for Pattern Recognition

Jun Hu, Huajin Tang, Kay Chen Tan, Haizhou Li, Luping Shi

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During the past few decades, remarkable progress has been made in solving pattern recognition problems using networks of spiking neurons. However, the issue of pattern recognition involving computational process from sensory encoding to synaptic learning remains underexplored, as most existing models or algorithms target only part of the computational process. Furthermore, many learning algorithms proposed in the literature neglect or pay little attention to sensory information encoding, which makes them incompatible with neural-realistic sensory signals encoded from real-world stimuli. By treating sensory coding and learning as a systematic process, we attempt to build an integrated model based on spiking neural networks (SNNs), which performs sensory neural encoding and supervised learning with precisely timed sequences of spikes. With emerging evidence of precise spike-timing neural activities, the view that information is represented by explicit firing times of action potentials rather than mean firing rates has been receiving increasing attention. The external sensory stimulation is first converted into spatiotemporal patterns using a latency-phase encoding method and subsequently transmitted to the consecutive network for learning. Spiking neurons are trained to reproduce target signals encoded with precisely timed spikes. We show that when a supervised spike-timing-based learning is used, different spatiotemporal patterns are recognized by different spike patterns with a high time precision in milliseconds.

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15. Supervised Learning in Multilayer Spiking Neural Networks

Ioana Sporea, André Grüning

https://doi.org/10.1162/neco_a_00396 | pp. 473-509

We introduce a supervised learning algorithm for multilayer spiking neural networks. The algorithm overcomes a limitation of existing learning algorithms: it can be applied to neurons firing multiple spikes in artificial neural networks with hidden layers. It can also, in principle, be used with any linearizable neuron model and allows different coding schemes of spike train patterns. The algorithm is applied successfully to classic linearly nonseparable benchmarks such as the XOR problem and the Iris data set, as well as to more complex classification and mapping problems. The algorithm has been successfully tested in the presence of noise, requires smaller networks than reservoir computing, and results in faster convergence than existing algorithms for similar tasks such as SpikeProp.

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16. Temporal Order Detection and Coding in Nervous Systems

Klaus M. Stiefel, Jonathan Tapson, André van Schaik

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This letter discusses temporal order coding and detection in nervous systems. Detection of temporal order in the external world is an adaptive function of nervous systems. In addition, coding based on the temporal order of signals can be used as an internal code. Such temporal order coding is a subset of temporal coding. We discuss two examples of processing the temporal order of external events: the auditory location detection system in birds and the visual direction detection system in flies. We then discuss how somatosensory stimulus intensities are translated into a temporal order code in the human peripheral nervous system. We next turn our attention to input order coding in the mammalian cortex. We review work demonstrating the capabilities of cortical neurons for detecting input order. We then discuss research refuting and demonstrating the representation of stimulus features in the cortex by means of input order. After some general theoretical considerations on input order detection and coding, we conclude by

discussing the existing and potential use of input order coding in neuromorphic engineering.

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17. The Kernel Semi–Least Squares Method for Sparse Distance Approximation

Samuel Epstein, Margrit Betke

https://doi.org/10.1162/neco_a_00401 | pp. 532-548

We extend the semi-least squares problem defined by Rao and Mitra (1971) to the kernel semi-least squares problem. We introduce subset projection, a technique that produces a solution to this problem. We show how the results of subset projection can be used to approximate a computationally expensive distance metric.

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18. Pavlov's Dog Associative Learning Demonstrated on Synaptic-Like Organic Transistors

O. Bichler, W. Zhao, F. Alibart, S. Pleutin, S. Lenfant, D. Vuillaume, C. Gamrat

https://doi.org/10.1162/neco_a_00377 | pp. 549-566

In this letter, we present an original demonstration of an associative learning neural network inspired by the famous Pavlov's dogs experiment. A single nanoparticle organic memory field effect transistor (NOMFET) is used to implement each synapse. We show how the physical properties of this dynamic memristive device can be used to perform low-power write operations for the learning and implement short-term association using temporal coding and spike-timing-dependent plasticity-based learning. An electronic circuit was built to validate the proposed learning scheme with packaged devices, with good reproducibility despite the complex synaptic-like dynamic of the NOMFET in pulse regime.

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19. Online Learning with (Multiple) Kernels: A Review

Tom Diethe, Mark Girolami

https://doi.org/10.1162/neco_a_00406 | pp. 567-625

This review examines kernel methods for online learning, in particular, multiclass classification. We examine margin-based approaches, stemming from Rosenblatt's original perceptron algorithm, as well as nonparametric probabilistic approaches that are based on the popular gaussian process framework. We also examine approaches to online learning that use combinations of kernels--online multiple kernel learning. We present empirical validation of a wide range of methods on a protein fold recognition data set, where different biological feature types are available, and two object recognition data sets, Caltech101 and Caltech256, where multiple feature spaces are available in terms of different image feature extraction methods.

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20. Opening the Black Box: Low-Dimensional Dynamics in High-Dimensional Recurrent Neural Networks

David Sussillo, Omri Barak

https://doi.org/10.1162/neco_a_00409 | pp. 626-649

Recurrent neural networks (RNNs) are useful tools for learning nonlinear relationships between time-varying inputs and outputs with complex temporal dependencies. Recently developed algorithms have been successful at training RNNs to perform a wide variety of tasks, but the resulting networks have been treated as black boxes: their mechanism of operation remains unknown. Here we explore the hypothesis that fixed points, both stable and unstable, and the linearized dynamics around them, can reveal crucial aspects of how RNNs implement their computations. Further, we explore the utility of linearization in areas of phase space that are not true fixed points but merely points of very slow movement. We present a simple optimization technique that is applied to trained RNNs to find the fixed and slow points of their dynamics. Linearization around these slow regions can be used to explore, or reverse-engineer, the behavior of the RNN. We describe the technique, illustrate it using simple examples, and finally showcase it on three high-dimensional RNN examples: a 3-bit flip-flop device, an input-dependent sine wave generator, and a two-point moving average. In all cases, the mechanisms of trained networks could be inferred from the sets of fixed and slow points and the linearized dynamics around them.

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21. System Identification of mGluR-Dependent Long-Term Depression

Tim Tambuyzer, Tariq Ahmed, C. James Taylor, Daniël Berckmans, Detlef Balschun, Jean■Marie Aerts

https://doi.org/10.1162/neco_a_00408 | pp. 650-670

Recent advances have started to uncover the underlying mechanisms of metabotropic glutamate receptor (mGluR)-dependent long-term depression (LTD). However, it is not completely clear how these mechanisms are linked, and it is believed that several crucial mechanisms remain to be revealed. In this study, we investigated whether system identification (SI) methods can be used to gain insight into the mechanisms of synaptic plasticity. SI methods have been shown to be an objective and powerful approach for describing how sensory neurons encode information about stimuli. However, to our knowledge, it is the first time that SI methods have been applied to electrophysiological brain slice recordings of synaptic plasticity responses. The results indicate that the SI approach is a valuable tool for reverse-engineering of mGluR-LTD responses. We suggest that such SI methods can aid in unraveling the complexities of synaptic function.

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22. Echo State Property Linked to an Input: Exploring a Fundamental Characteristic of Recurrent Neural Networks

G. Manjunath, Herbert Jaeger

https://doi.org/10.1162/neco_a_00411 | pp. 671-696

The echo state property is a key for the design and training of recurrent neural networks within the paradigm of reservoir computing. In intuitive terms, this is a passivity condition: a network having this property, when driven by an input signal, will become entrained by the input and develop an internal response signal. This excited internal dynamics can be seen as a high-dimensional, nonlinear, unique transform of the input with a rich memory content. This view has implications for understanding neural dynamics beyond the field of reservoir computing. Available definitions and theorems concerning the echo state property, however, are of little practical use because they do not relate the network response to temporal or statistical properties of the

driving input. Here we present a new definition of the echo state property that directly connects it to such properties. We derive a fundamental 0-1 law: if the input comes from an ergodic source, the network response has the echo state property with probability one or zero, independent of the given network. Furthermore, we give a sufficient condition for the echo state property that connects statistical characteristics of the input to algebraic properties of the network connection matrix. The mathematical methods that we employ are freshly imported from the young field of nonautonomous dynamical systems theory. Since these methods are not yet well known in neural computation research, we introduce them in some detail. As a side story, we hope to demonstrate the eminent usefulness of these methods.

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23. Movement Duration, Fitts's Law, and an Infinite-Horizon Optimal Feedback Control Model for Biological Motor Systems

Ning Qian, Yu Jiang, Zhong■Ping Jiang, Pietro Mazzoni

https://doi.org/10.1162/neco_a_00410 | pp. 697-724

Optimization models explain many aspects of biological goal-directed movements. However, most such models use a finite-horizon formulation, which requires a prefixed movement duration to define a cost function and solve the optimization problem. To predict movement duration, these models have to be run multiple times with different prefixed durations until an appropriate duration is found by trial and error. The constrained minimum time model directly predicts movement duration; however, it does not consider sensory feedback and is thus applicable only to open-loop movements. To address these problems, we analyzed and simulated an infinite-horizon optimal feedback control model, with linear plants, that contains both control-dependent and control-independent noise and optimizes steady-state accuracy and energetic costs per unit time. The model applies the steady-state estimator and controller continuously to guide an effector to, and keep it at, target position. As such, it integrates movement control and posture maintenance without artificially dividing them with a precise, prefixed time boundary. Movement pace is determined by the model parameters, and the duration is an emergent property with trial-to-trial variability. By considering the mean duration, we derived both the log and power forms of Fitts's law as different approximations of the model. Moreover, the model reproduces typically observed velocity profiles and occasional transient overshoots. For unbiased sensory feedback, the effector reaches the target without bias, in contrast to finite-horizon models that systematically undershoot target when energetic cost is considered. Finally, the model does not involve backward and forward sweeps in time, its stability is easily checked, and the same solution applies to movements of different initial conditions and distances. We argue that biological systems could use steady-state solutions as default control mechanisms and might seek additional optimization of transient costs when justified or demanded by task or context.

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24. Sufficient Dimension Reduction via Squared-Loss Mutual Information Estimation

Taiji Suzuki, Masashi Sugiyama

https://doi.org/10.1162/neco_a_00407 | pp. 725-758

The goal of sufficient dimension reduction in supervised learning is to find the low-dimensional subspace of input features that contains all of the information about the output values that the input features possess. In this letter, we propose a novel sufficient dimension-reduction method using a squared-loss variant of mutual

information as a dependency measure. We apply a density-ratio estimator for approximating squared-loss mutual information that is formulated as a minimum contrast estimator on parametric or nonparametric models. Since cross-validation is available for choosing an appropriate model, our method does not require any prespecified structure on the underlying distributions. We elucidate the asymptotic bias of our estimator on parametric models and the asymptotic convergence rate on nonparametric models. The convergence analysis utilizes the uniform tail-bound of a U-process, and the convergence rate is characterized by the bracketing entropy of the model. We then develop a natural gradient algorithm on the Grassmann manifold for sufficient subspace search. The analytic formula of our estimator allows us to compute the gradient efficiently. Numerical experiments show that the proposed method compares favorably with existing dimension-reduction approaches on artificial and benchmark data sets.

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25. A Unified Classification Model Based on Robust Optimization

Akiko Takeda, Hiroyuki Mitsugi, Takafumi Kanamori

https://doi.org/10.1162/neco_a_00412 | pp. 759-804

A wide variety of machine learning algorithms such as the support vector machine (SVM), minimax probability machine (MPM), and Fisher discriminant analysis (FDA) exist for binary classification. The purpose of this letter is to provide a unified classification model that includes these models through a robust optimization approach. This unified model has several benefits. One is that the extensions and improvements intended for SVMs become applicable to MPM and FDA, and vice versa. For example, we can obtain nonconvex variants of MPM and FDA by mimicking Perez-Cruz, Weston, Hermann, and Schölkopf's (2003) extension from convex ν-SVM to nonconvex Eν-SVM. Another benefit is to provide theoretical results concerning these learning methods at once by dealing with the unified model. We give a statistical interpretation of the unified classification model and prove that the model is a good approximation for the worst-case minimization of an expected loss with respect to the uncertain probability distribution. We also propose a nonconvex optimization algorithm that can be applied to nonconvex variants of existing learning methods and show promising numerical results.

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26. Enhanced Gradient for Training Restricted Boltzmann Machines

Kyunghyun Cho, Tapani Raiko, Alexander Ilin

https://doi.org/10.1162/neco_a_00397 | pp. 805-831

Restricted Boltzmann machines (RBMs) are often used as building blocks in greedy learning of deep networks. However, training this simple model can be laborious. Traditional learning algorithms often converge only with the right choice of metaparameters that specify, for example, learning rate scheduling and the scale of the initial weights. They are also sensitive to specific data representation. An equivalent RBM can be obtained by flipping some bits and changing the weights and biases accordingly, but traditional learning rules are not invariant to such transformations. Without careful tuning of these training settings, traditional algorithms can easily get stuck or even diverge. In this letter, we present an enhanced gradient that is derived to be invariant to bit-flipping transformations. We experimentally show that the enhanced gradient yields more stable training of RBMs both when used with a fixed learning rate and an adaptive one.

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27. Generalization and Multirate Models of Motor Adaptation

Hirokazu Tanaka, John W. Krakauer, Terrence J. Sejnowski

https://doi.org/10.1162/neco_a_00262 | pp. 939-966

When subjects adapt their reaching movements in the setting of a systematic force or visual perturbation, generalization of adaptation can be assessed psychophysically in two ways: by testing untrained locations in the work space at the end of adaptation (slow postadaptation generalization) or by determining the influence of an error on the next trial during adaptation (fast trial-by-trial generalization). These two measures of generalization have been widely used in psychophysical studies, but the reason that they might differ has not been addressed explicitly. Our goal was to develop a computational framework for determining when a two-state model is justified by the data and to explore the implications of these two types of generalization for neural representations of movements. We first investigated, for single-target learning, how well standard statistical model selection procedures can discriminate two-process models from single-process models when learning and retention coefficients were systematically varied. We then built a two-state model for multitarget learning and showed that if an adaptation process is indeed two-rate, then the postadaptation generalization approach primarily probes the slow process, whereas the trial-by-trial generalization approach is most informative about the fast process. The fast process, due to its strong sensitivity to trial error, contributes predominantly to trial-by-trial generalization, whereas the strong retention of the slow system contributes predominantly to postadaptation generalization. Thus, when adaptation can be shown to be two-rate, the two measures of generalization may probe different brain representations of movement direction.

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28. The Computational Power of Interactive Recurrent Neural Networks

Jérémie Cabessa, Hava T. Siegelmann

https://doi.org/10.1162/neco_a_00263 | pp. 996-1019

In classical computation, rational- and real-weighted recurrent neural networks were shown to be respectively equivalent to and strictly more powerful than the standard Turing machine model. Here, we study the computational power of recurrent neural networks in a more biologically oriented computational framework, capturing the aspects of sequential interactivity and persistence of memory. In this context, we prove that so-called interactive rational- and real-weighted neural networks show the same computational powers as interactive Turing machines and interactive Turing machines with advice, respectively. A mathematical characterization of each of these computational powers is also provided. It follows from these results that interactive real-weighted neural networks can perform uncountably many more translations of information than interactive Turing machines, making them capable of super-Turing capabilities.

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29. Erratum (“A State-Space Analysis for Reconstruction of Goal-Directed Movements Using Neural Signals” Neural Computation, Volume 18, Number 10, pp. 2465–2494)

Iahn Cajigas, Lakshminarayan Srinivasan

https://doi.org/10.1162/neco_c_00279 | pp. 1106-1107

The purpose of this note is to address a typo present in the last term of Equation 2.16 of Srinivasan et al. (2006) 2.16Equation 2.16 should have instead been where the corrected term from Equation 2.16 has been outlined.

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30. Stochastic Perturbation Methods for Spike-Timing-Dependent Plasticity

Todd K. Leen, Robert Friel

https://doi.org/10.1162/neco_a_00267 | pp. 1109-1146

Online machine learning rules and many biological spike-timing-dependent plasticity (STDP) learning rules generate jump process Markov chains for the synaptic weights. We give a perturbation expansion for the dynamics that, unlike the usual approximation by a Fokker-Planck equation (FPE), is well justified. Our approach extends the related system size expansion by giving an expansion for the probability density as well as its moments. We apply the approach to two observed STDP learning rules and show that in regimes where the FPE breaks down, the new perturbation expansion agrees well with Monte Carlo simulations. The methods are also applicable to the dynamics of stochastic neural activity. Like previous ensemble analyses of STDP, we focus on equilibrium solutions, although the methods can in principle be applied to transients as well.

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31. Dynamical Synapses Enhance Neural Information Processing: Gracefulness, Accuracy, and Mobility

Chi Chung Alan Fung, K. Y. Michael Wong, He Wang, Si Wu

https://doi.org/10.1162/neco_a_00269 | pp. 1147-1185

Experimental data have revealed that neuronal connection efficacy exhibits two forms of short-term plasticity: short-term depression (STD) and short-term facilitation (STF). They have time constants residing between fast neural signaling and rapid learning and may serve as substrates for neural systems manipulating temporal information on relevant timescales. This study investigates the impact of STD and STF on the dynamics of continuous attractor neural networks and their potential roles in neural information processing. We find that STD endows the network with slow-decaying plateau behaviors: the network that is initially being stimulated to an active state decays to a silent state very slowly on the timescale of STD rather than on that of neuralsignaling. This provides a mechanism for neural systems to hold sensory memory easily and shut off persistent activities gracefully. With STF, we find that the network can hold a memory trace of external inputs in the facilitated neuronal interactions, which provides a way to stabilize the network response to noisy inputs, leading to improved accuracy in population decoding. Furthermore, we find that STD increases the mobility of the network states. The increased mobility enhances the tracking performance of the network in response to time-varying stimuli, leading to anticipative neural responses. In general, we find that STD and STP tend to have opposite effects on network dynamics and complementary computational advantages, suggesting that the brain may employ a strategy of weighting them differentially depending on the computational purpose.

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32. Reinforcement-Based Decision Making in Corticostriatal Circuits: Mutual Constraints by Neurocomputational and Diffusion Models

Roger Ratcliff, Michael J. Frank

https://doi.org/10.1162/neco_a_00270 | pp. 1186-1229

In this letter, we examine the computational mechanisms of reinforce-ment-based decision making. We bridge the gap across multiple levels of analysis, from neural models of corticostriatal circuits-the basal ganglia (BG) model (Frank, 2005 , 2006) to simpler but mathematically tractable diffusion models of two-choice decision making. Specifically, we generated simulated data from the BG model and fit the diffusion model (Ratcliff, 1978) to it. The standard diffusion model fits underestimated response times under conditions of high response and reinforcement conflict. Follow-up fits showed good fits to the data both by increasing nondecision time and by raising decision thresholds as a function of conflict and by allowing this threshold to collapse with time. This profile captures the role and dynamics of the subthalamic nucleus in BG circuitry, and as such, parametric modulations of projection strengths from this nucleus were associated with parametric increases in decision boundary and its modulation by conflict. We then present data from a human reinforcement learning experiment involving decisions with low- and high-reinforcement conflict. Again, the standard model failed to fit the data, but we found that two variants similar to those that fit the BG model data fit the experimental data, thereby providing a convergence of theoretical accounts of complex interactive decision-making mechanisms consistent with available data. This work also demonstrates how to make modest modifications to diffusion models to summarize core computations of the BG model. The result is a better fit and understanding of reinforcement-based choice data than that which would have occurred with either model alone.

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33. Statistical Mechanics of Reward-Modulated Learning in Decision-Making Networks

Kentaro Katahira, Kazuo Okanoya, Masato Okada

https://doi.org/10.1162/neco_a_00264 | pp. 1230-1270

The neural substrates of decision making have been intensively studied using experimental and computational approaches. Alternative-choice tasks accompanying reinforcement have often been employed in investigations into decision making. Choice behavior has been empirically found in many experiments to follow Herrnstein's matching law. A number of theoretical studies have been done on explaining the mechanisms responsible for matching behavior. Various learning rules have been proved in these studies to achieve matching behavior as a steady state of learning processes. The models in the studies have consisted of a few parameters. However, a large number of neurons and synapses are expected to participate in decision making in the brain. We investigated learning behavior in simple but large-scale decision-making networks. We considered the covariance learning rule, which has been demonstrated to achieve matching behavior as a steady state (Loewenstein & Seung, 2006). We analyzed model behavior in a thermodynamic limit where the number of plastic synapses went to infinity. By means of techniques of the statistical mechanics, we can derive deterministic differential equations in this limit for the order parameters, which allow an exact calculation of the evolution of choice behavior. As a result, we found that matching behavior cannot be a steady state of learning when the fluctuations in input from individual sensory neurons are so large that they affect the net input to value-encoding neurons. This situation naturally arises when the synaptic strength is sufficiently strong and the excitatory input and the inhibitory

input to the value-encoding neurons are balanced. The deviation from matching behavior is caused by increasing variance in the input potential due to the diffusion of synaptic efficacies. This effect causes an undermatching phenomenon, which has been often observed in behavioral experiments.

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34. Learning Invariance from Natural Images Inspired by Observations in the Primary Visual Cortex

Michael Teichmann, Jan Wiltschut, Fred H. Hamker

https://doi.org/10.1162/neco_a_00268 | pp. 1271-1296

The human visual system has the remarkable ability to largely recognize objects invariant of their position, rotation, and scale. A good interpretation of neurobiological findings involves a computational model that simulates signal processing of the visual cortex. In part, this is likely achieved step by step from early to late areas of visual perception. While several algorithms have been proposed for learning feature detectors, only few studies at hand cover the issue of biologically plausible learning of such invariance. In this study, a set of Hebbian learning rules based on calcium dynamics and homeostatic regulations of single neurons is proposed. Their performance is verified within a simple model of the primary visual cortex to learn so-called complex cells, based on a sequence of static images. As a result, the learned complex-cell responses are largely invariant to phase and position.

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35. Reduction from Cost-Sensitive Ordinal Ranking to Weighted Binary Classification

Hsuan-Tien Lin, Ling Li

https://doi.org/10.1162/neco_a_00265 | pp. 1329-1367

We present a reduction framework from ordinal ranking to binary classification. The framework consists of three steps: extracting extended examples from the original examples, learning a binary classifier on the extended examples with any binary classification algorithm, and constructing a ranker from the binary classifier. Based on the framework, we show that a weighted 0/1 loss of the binary classifier upper-bounds the mislabeling cost of the ranker, both error-wise and regret-wise. Our framework allows not only the design of good ordinal ranking algorithms based on well-tuned binary classification approaches, but also the derivation of new generalization bounds for ordinal ranking from known bounds for binary classification. In addition, our framework unifies many existing ordinal ranking algorithms, such as perceptron ranking and support vector ordinal regression. When compared empirically on benchmark data sets, some of our newly designed algorithms enjoy advantages in terms of both training speed and generalization performance over existing algorithms. In addition, the newly designed algorithms lead to better cost-sensitive ordinal ranking performance, as well as improved listwise ranking performance.

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36. Entropy Estimation in Turing's Perspective

Zhiyi Zhang

https://doi.org/10.1162/neco_a_00266 | pp. 1368-1389

A new nonparametric estimator of Shannon's entropy on a countable alphabet is proposed and analyzed against the well-known plug-in estimator. The proposed estimator is developed based on Turing's formula, which recovers distributional characteristics on the subset of the alphabet not covered by a size-n sample. The fundamental switch in perspective brings about substantial gain in estimation accuracy for every distribution with finite entropy. In general, a uniform variance upper bound is established for the entire class of distributions with finite entropy that decays at a rate of O(ln(n)/n) compared to O([ln(n)]2/n) for the plug-in. In a wide range of subclasses, the variance of the proposed estimator converges at a rate of O(1/n), and this rate of convergence carries over to the convergence rates in mean squared errors in many subclasses. Specifically, for any finite alphabet, the proposed estimator has a bias decaying exponentially in n. Several new bias-adjusted estimators are also discussed.

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37. A Proof of Convergence of the Concave-Convex Procedure Using Zangwill's Theory

Bharath K. Sriperumbudur, Gert Lanckriet

https://doi.org/10.1162/neco_a_00283 | pp. 1391-1407

The concave-convex procedure (CCCP) is an iterative algorithm that solves d.c. (difference of convex functions) programs as a sequence of convex programs. In machine learning, CCCP is extensively used in many learning algorithms, including sparse support vector machines (SVMs), transductive SVMs, and sparse principal component analysis. Though CCCP is widely used in many applications, its convergence behavior has not gotten a lot of specific attention. Yuille and Rangarajan analyzed its convergence in their original paper; however, we believe the analysis is not complete. The convergence of CCCP can be derived from the convergence of the d.c. algorithm (DCA), proposed in the global optimization literature to solve general d.c. programs, whose proof relies on d.c. duality. In this note, we follow a different reasoning and show how Zangwill's global convergence theory of iterative algorithms provides a natural framework to prove the convergence of CCCP. This underlines Zangwill's theory as a powerful and general framework to deal with the convergence issues of iterative algorithms, after also being used to prove the convergence of algorithms like expectation-maximization and generalized alternating minimization. In this note, we provide a rigorous analysis of the convergence of CCCP by addressing two questions: When does CCCP find a local minimum or a stationary point of the d.c. program under consideration? and when does the sequence generated by CCCP converge? We also present an open problem on the issue of local convergence of CCCP.

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38. On the Relation Between Encoding and Decoding of Neuronal Spikes

Shinsuke Koyama

https://doi.org/10.1162/neco_a_00279 | pp. 1408-1425

Neural coding is a field of study that concerns how sensory information is represented in the brain by networks of neurons. The link between external stimulus and neural response can be studied from two parallel points of view. The first, neural encoding, refers to the mapping from stimulus to response. It focuses primarily on understanding how neurons respond to a wide variety of stimuli and constructing models that accurately describe the stimulus-response relationship. Neural decoding refers to the reverse mapping, from response to stimulus, where the challenge is to reconstruct a stimulus from the spikes it evokes. Since

neuronal response is stochastic, a one-to-one mapping of stimuli into neural responses does not exist, causing a mismatch between the two viewpoints of neural coding. Here we use these two perspectives to investigate the question of what rate coding is, in the simple setting of a single stationary stimulus parameter and a single stationary spike train represented by a renewal process. We show that when rate codes are defined in terms of encoding, that is, the stimulus parameter is mapped onto the mean firing rate, the rate decoder given by spike counts or the sample mean does not always efficiently decode the rate codes, but it can improve efficiency in reading certain rate codes when correlations within a spike train are taken into account.

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39. Analytical Integrate-and-Fire Neuron Models with Conductance-Based Dynamics and Realistic Postsynaptic Potential Time Course for Event-Driven Simulation Strategies

Michelle Rudolph-Lilith, Mathieu Dubois, Alain Destexhe

https://doi.org/10.1162/neco_a_00278 | pp. 1426-1461

In a previous paper (Rudolph & Destexhe, 2006), we proposed various models, the gIF neuron models, of analytical integrate-and-fire (IF) neurons with conductance-based (COBA) dynamics for use in event-driven simulations. These models are based on an analytical approximation of the differential equation describing the IF neuron with exponential synaptic conductances and were successfully tested with respect to their response to random and oscillating inputs. Because they are analytical and mathematically simple, the gIF models are best suited for fast event-driven simulation strategies. However, the drawback of such models is they rely on a nonrealistic postsynaptic potential (PSP) time course, consisting of a discontinuous jump followed by a decay governed by the membrane time constant. Here, we address this limitation by conceiving an analytical approximation of the COBA IF neuron model with the full PSP time course. The subthreshold and suprathreshold response of this gIF4 model reproduces remarkably well the postsynaptic responses of the numerically solved passive membrane equation subject to conductance noise, while gaining at least two orders of magnitude in computational performance. Although the analytical structure of the gIF4 model is more complex than that of its predecessors due to the necessity of calculating future spike times, a simple and fast algorithmic implementation for use in large-scale neural network simulations is proposed.

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40. Markov Chain Monte Carlo Methods for State-Space Models with Point Process Observations

Ke Yuan, Mark Girolami, Mahesan Niranjan

https://doi.org/10.1162/neco_a_00281 | pp. 1462-1486

This letter considers how a number of modern Markov chain Monte Carlo (MCMC) methods can be applied for parameter estimation and inference in state-space models with point process observations. We quantified the efficiencies of these MCMC methods on synthetic data, and our results suggest that the Reimannian manifold Hamiltonian Monte Carlo method offers the best performance. We further compared such a method with a previously tested variational Bayes method on two experimental data sets. Results indicate similar performance on the large data sets and superior performance on small ones. The work offers an extensive suite of MCMC algorithms evaluated on an important class of models for physiological

signal analysis.

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41. Adaptive Optimal Control Without Weight Transport

Lakshminarayan V. Chinta, Douglas Tweed
https://doi.org/10.1162/neco_a_00277 | pp. 1487-1518

Many neural control systems are at least roughly optimized, but how is optimal control learned? There are algorithms for this purpose, but in their current forms, they are not suited for biological neural networks because they rely on a type of communication that is not available in the brain, namely, weight transport-transmitting the strengths, or "weights," of individual synapses to other synapses and neurons. Here we show how optimal control can be learned without weight transport. Our method involves a set of simple mechanisms that can compensate for the absence of weight transport in the brain and so may be useful for neural computation generally.

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42. The Neural Representation of Time: An Information-Theoretic Perspective

Joachim Haß, J. Michael Herrmann
https://doi.org/10.1162/neco_a_00280 | pp. 1519-1552

A prominent finding in psychophysical experiments on time perception is Weber's law, the linear scaling of timing errors with duration. The ability to reproduce this scaling has been taken as a criterion for the validity of neurocomputational models of time perception. However, the origin of Weber's law remains unknown, and currently only a few models generically reproduce it. Here, we use an information-theoretical framework that considers the neuronal mechanisms of time perception as stochastic processes to investigate the statistical origin of Weber's law in time perception and also its frequently observed deviations. Under the assumption that the brain is able to compute optimal estimates of time, we find that Weber's law only holds exactly if the estimate is based on temporal changes in the variance of the process. In contrast, the timing errors scale sublinearly with time if the systematic changes in the mean of a process are used for estimation, as is the case in the majority of time perception models, while estimates based on temporal correlations result in a superlinear scaling. This hierarchy of temporal information is preserved if several sources of temporal information are available. Furthermore, we consider the case of multiple stochastic processes and study the examples of a covariance-based model and a model based on synfire chains. This approach reveals that existing neurocomputational models of time perception can be classified as mean-, variance- and correlation-based processes and allows predictions about the scaling of the resulting timing errors.

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43. The Successor Representation and Temporal Context

Samuel J. Gershman, Christopher D. Moore, Michael T. Todd, Kenneth A. Norman, Per B. Sederberg
https://doi.org/10.1162/neco_a_00282 | pp. 1553-1568

The successor representation was introduced into reinforcement learning by Dayan (1993) as a means of facilitating generalization between states with similar successors. Although reinforcement learning in general has been used extensively as a model of psychological and neural processes, the psychological validity of the successor representation has yet to be explored. An interesting possibility is that the successor

representation can be used not only for reinforcement learning but for episodic learning as well. Our main contribution is to show that a variant of the temporal context model (TCM; Howard & Kahana, 2002), an influential model of episodic memory, can be understood as directly estimating the successor representation using the temporal difference learning algorithm (Sutton & Barto, 1998). This insight leads to a generalization of TCM and new experimental predictions. In addition to casting a new normative light on TCM, this equivalence suggests a previously unexplored point of contact between different learning systems.

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44. Learning Coefficient of Generalization Error in Bayesian Estimation and Vandermonde Matrix-Type Singularity

Miki Aoyagi, Kenji Nagata

https://doi.org/10.1162/neco_a_00271 | pp. 1569-1610

The term algebraic statistics arises from the study of probabilistic models and techniques for statistical inference using methods from algebra and geometry (Sturmfels, 2009). The purpose of our study is to consider the generalization error and stochastic complexity in learning theory by using the log-canonical threshold in algebraic geometry. Such thresholds correspond to the main term of the generalization error in Bayesian estimation, which is called a learning coefficient (Watanabe, 2001a , 2001b). The learning coefficient serves to measure the learning efficiencies in hierarchical learning models. In this letter, we consider learning coefficients for Vandermonde matrix-type singularities, by using a new approach: focusing on the generators of the ideal, which defines singularities. We give tight new bound values of learning coefficients for the Vandermonde matrix-type singularities and the explicit values with certain conditions. By applying our results, we can show the learning coefficients of three-layered neural networks and normal mixture models.

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45. Causality, Conditional Independence, and Graphical Separation in Settable Systems

Karim Chalak, Halbert White

https://doi.org/10.1162/neco_a_00295 | pp. 1611-1668

We study the connections between causal relations and conditional independence within the settable systems extension of the Pearl causal model (PCM). Our analysis clearly distinguishes between causal notions and probabilistic notions, and it does not formally rely on graphical representations. As a foundation, we provide definitions in terms of suitable functional dependence for direct causality and for indirect and total causality via and exclusive of a set of variables. Based on these foundations, we provide causal and stochastic conditions formally characterizing conditional dependence among random vectors of interest in structural systems by stating and proving the conditional Reichenbach principle of common cause, obtaining the classical Reichenbach principle as a corollary. We apply the conditional Reichenbach principle to show that the useful tools of d-separation and D-separation can be employed to establish conditional independence within suitably restricted settable systems analogous to Markovian PCMs.

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46. Dynamic State and Parameter Estimation Applied to Neuromorphic Systems

Emre Neftci, Bryan Toth, Giacomo Indiveri, Henry D. I. Abarbanel

https://doi.org/10.1162/neco_a_00293 | pp. 1669-1694

Neuroscientists often propose detailed computational models to probe the properties of the neural systems they study. With the advent of neuromorphic engineering, there is an increasing number of hardware electronic analogs of biological neural systems being proposed as well. However, for both biological and hardware systems, it is often difficult to estimate the parameters of the model so that they are meaningful to the experimental system under study, especially when these models involve a large number of states and parameters that cannot be simultaneously measured. We have developed a procedure to solve this problem in the context of interacting neural populations using a recently developed dynamic state and parameter estimation (DSPE) technique. This technique uses synchronization as a tool for dynamically coupling experimentally measured data to its corresponding model to determine its parameters and internal state variables. Typically experimental data are obtained from the biological neural system and the model is simulated in software; here we show that this technique is also efficient in validating proposed network models for neuromorphic spike-based very large-scale integration (VLSI) chips and that it is able to systematically extract network parameters such as synaptic weights, time constants, and other variables that are not accessible by direct observation. Our results suggest that this method can become a very useful tool for model-based identification and configuration of neuromorphic multichip VLSI systems.

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47. Neural Information Processing with Feedback Modulations

Wenhao Zhang, Si Wu

https://doi.org/10.1162/neco_a_00296 | pp. 1695-1721

Descending feedback connections, together with ascending feedforward ones, are the indispensable parts of the sensory pathways in the central nervous system. This study investigates the potential roles of feedback interactions in neural information processing. We consider a two-layer continuous attractor neural network (CANN), in which neurons in the first layer receive feedback inputs from those in the second one. By utilizing the intrinsic property of a CANN, we use a projection method to reduce the dimensionality of the network dynamics significantly. The simplified dynamics allows us to elucidate the effects of feedback modulation analytically. We find that positive feedback enhances the stability of the network state, leading to an improved population decoding performance, whereas negative feedback increases the mobility of the network state, inducing spontaneously moving bumps. For strong, negative feedback interaction, the network response to a moving stimulus can lead the actual stimulus position, achieving an anticipative behavior. The biological implications of these findings are discussed. The simulation results agree well with our theoretical analysis.

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48. Identification of Directed Influence: Granger Causality, Kullback-Leibler Divergence, and Complexity

Abd■Krim Seghouane, ■■■-■■■ ■■■■■

https://doi.org/10.1162/neco_a_00291 | pp. 1722-1739

Detecting and characterizing causal interdependencies and couplings between different activated brain areas from functional neuroimage time series measurements of their activity constitutes a significant step toward understanding the process of brain functions. In this letter, we make the simple point that all current statistics used to make inferences about directed influences in functional neuroimage time series are variants of the same underlying quantity. This includes directed transfer entropy, transinformation, Kullback-Leibler formulations, conditional mutual information, and Granger causality. Crucially, in the case of autoregressive modeling, the underlying quantity is the likelihood ratio that compares models with and without directed influences from the past when modeling the influence of one time series on another. This framework is also used to derive the relation between these measures of directed influence and the complexity or the order of directed influence. These results provide a framework for unifying the Kullback-Leibler divergence, Granger causality, and the complexity of directed influence.

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49. Fisher and Shannon Information in Finite Neural Populations

Stuart Yarrow, Edward Challis, Peggy Seriès

https://doi.org/10.1162/neco_a_00292 | pp. 1740-1780

The precision of the neural code is commonly investigated using two families of statistical measures: Shannon mutual information and derived quantities when investigating very small populations of neurons and Fisher information when studying large populations. These statistical tools are no longer the preserve of theorists and are being applied by experimental research groups in the analysis of empirical data. Although the relationship between information-theoretic and Fisher-based measures in the limit of infinite populations is relatively well understood, how these measures compare in finite-size populations has not yet been systematically explored. We aim to close this gap. We are particularly interested in understanding which stimuli are best encoded by a given neuron within a population and how this depends on the chosen measure. We use a novel Monte Carlo approach to compute a stimulus-specific decomposition of the mutual information (the SSI) for populations of up to 256 neurons and show that Fisher information can be used to accurately estimate both mutual information and SSI for populations of the order of 100 neurons, even in the presence of biologically realistic variability, noise correlations, and experimentally relevant integration times. According to both measures, the stimuli that are best encoded are those falling at the flanks of the neuron's tuning curve. In populations of fewer than around 50 neurons, however, Fisher information can be misleading.

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50. Variation in the Local Motion Statistics of Real-Life Optic Flow Scenes

Szonya Durant, Johannes M. Zanker

https://doi.org/10.1162/neco_a_00294 | pp. 1781-1805

Optic flow motion patterns can be a rich source of information about our own movement and about the structure of the environment we are moving in. We investigate the information available to the brain under real operating conditions by analyzing video sequences generated by physically moving a camera through various typical human environments. We consider to what extent the motion signal maps generated by a biologically plausible, two-dimensional array of correlation-based motion detectors (2DMD) not only depend on egomotion, but also reflect the spatial setup of such environments. We analyzed the local motion outputs by extracting the relative amounts of detected directions and comparing the spatial distribution of the motion

signals to that of idealized optic flow. Using a simple template matching estimation technique, we are able to extract the focus of expansion and find relatively small errors that are distributed in characteristic patterns in different scenes. This shows that all types of scenes provide suitable motion information for extracting ego motion despite the substantial levels of noise affecting the motion signal distributions, attributed to the sparse nature of optic flow and the presence of camera jitter. However, there are large differences in the shape of the direction distributions between different types of scenes; in particular, man-made office scenes are heavily dominated by directions in the cardinal axes, which is much less apparent in outdoor forest scenes. Further examination of motion magnitudes at different scales and the location of motion information in a scene revealed different patterns across different scene categories. This suggests that self-motion patterns are not only relevant for deducing heading direction and speed but also provide a rich information source for scene structure and could be important for the rapid formation of the gist of a scene under normal human locomotion.

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51. A Computational Feedforward Model Predicts Categorization of Masked Emotional Body Language for Longer, but Not for Shorter, Latencies

Bernard M. C. Stienen, Konrad Schindler, Béatrice de Gelder

https://doi.org/10.1162/neco_a_00305 | pp. 1806-1821

Given the presence of massive feedback loops in brain networks, it is difficult to disentangle the contribution of feedforward and feedback processing to the recognition of visual stimuli, in this case, of emotional body expressions. The aim of the work presented in this letter is to shed light on how well feedforward processing explains rapid categorization of this important class of stimuli. By means of parametric masking, it may be possible to control the contribution of feedback activity in human participants. A close comparison is presented between human recognition performance and the performance of a computational neural model that exclusively modeled feedforward processing and was engineered to fulfill the computational requirements of recognition. Results show that the longer the stimulus onset asynchrony (SOA), the closer the performance of the human participants was to the values predicted by the model, with an optimum at an SOA of 100 ms. At short SOA latencies, human performance deteriorated, but the categorization of the emotional expressions was still above baseline. The data suggest that, although theoretically, feedback arising from inferotemporal cortex is likely to be blocked when the SOA is 100 ms, human participants still seem to rely on more local visual feedback processing to equal the model's performance.

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52. Simple Deterministically Constructed Cycle Reservoirs with Regular Jumps

Ali Rodan, Peter Ti■o

https://doi.org/10.1162/neco_a_00297 | pp. 1822-1852

A new class of state-space models, reservoir models, with a fixed state transition structure (the "reservoir") and an adaptable readout from the state space, has recently emerged as a way for time series processing and modeling. Echo state network (ESN) is one of the simplest, yet powerful, reservoir models. ESN models are generally constructed in a randomized manner. In our previous study (Rodan & Ti■o, 2011), we showed that a very simple, cyclic, deterministically generated reservoir can yield performance competitive with standard ESN. In this contribution, we extend our previous study in three aspects. First, we introduce a

novel simple deterministic reservoir model, cycle reservoir with jumps (CRJ), with highly constrained weight values, that has superior performance to standard ESN on a variety of temporal tasks of different origin and characteristics. Second, we elaborate on the possible link between reservoir characterizations, such as eigenvalue distribution of the reservoir matrix or pseudo-Lyapunov exponent of the input-driven reservoir dynamics, and the model performance. It has been suggested that a uniform coverage of the unit disk by such eigenvalues can lead to superior model performance. We show that despite highly constrained eigenvalue distribution, CRJ consistently outperforms ESN (which has much more uniform eigenvalue coverage of the unit disk). Also, unlike in the case of ESN, pseudo-Lyapunov exponents of the selected optimal CRJ models are consistently negative. Third, we present a new framework for determining the short-term memory capacity of linear reservoir models to a high degree of precision. Using the framework, we study the effect of shortcut connections in the CRJ reservoir topology on its memory capacity.

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53. Multiple Kernel Learning with Gaussianity Measures

Hideitsu Hino, Nima Reyhani, Noboru Murata

https://doi.org/10.1162/neco_a_00299 | pp. 1853-1881

Kernel methods are known to be effective for nonlinear multivariate analysis. One of the main issues in the practical use of kernel methods is the selection of kernel. There have been a lot of studies on kernel selection and kernel learning. Multiple kernel learning (MKL) is one of the promising kernel optimization approaches. Kernel methods are applied to various classifiers including Fisher discriminant analysis (FDA). FDA gives the Bayes optimal classification axis if the data distribution of each class in the feature space is a gaussian with a shared covariance structure. Based on this fact, an MKL framework based on the notion of gaussianity is proposed. As a concrete implementation, an empirical characteristic function is adopted to measure gaussianity in the feature space associated with a convex combination of kernel functions, and two MKL algorithms are derived. From experimental results on some data sets, we show that the proposed kernel learning followed by FDA offers strong classification power.

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54. Conditional Association

Sohan Seth, José C. Pr■■ncipe

https://doi.org/10.1162/neco_a_00298 | pp. 1882-1905

Estimating conditional dependence between two random variables given the knowledge of a third random variable is essential in neuroscientific applications to understand the causal architecture of a distributed network. However, existing methods of assessing conditional dependence, such as the conditional mutual information, are computationally expensive, involve free parameters, and are difficult to understand in the context of realizations. In this letter, we discuss a novel approach to this problem and develop a computationally simple and parameter-free estimator. The difference between the proposed approach and the existing ones is that the former expresses conditional dependence in terms of a finite set of realizations, whereas the latter use random variables, which are not available in practice. We call this approach conditional association, since it is based on a generalization of the concept of association to arbitrary metric spaces. We also discuss a novel and computationally efficient approach of generating surrogate data for evaluating the significance of the acquired association value.

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55. A Neurocomputational Approach to Prepositional Phrase Attachment Ambiguity Resolution

Kailash Nadh, Christian Huyck

https://doi.org/10.1162/neco_a_00290 | pp. 1906-1925

A neurocomputational model based on emergent massively overlapping neural cell assemblies (CAs) for resolving prepositional phrase (PP) attachment ambiguity is described. PP attachment ambiguity is a well-studied task in natural language processing and is a case where semantics is used to determine the syntactic structure. A large network of biologically plausible fatiguing leaky integrate-and-fire neurons is trained with semantic hierarchies (obtained from WordNet) on sentences with PP attachment ambiguity extracted from the Penn Treebank corpus. During training, overlapping CAs representing semantic similarities between the component words of the ambiguous sentences emerge and then act as categorizers for novel input. The resulting average resolution accuracy of 84.56% is on par with known machine learning algorithms.

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56. Improved Generative Semisupervised Learning Based on Finely Grained Component-Conditional Class Labeling

David J. Miller, Jayaram Raghuram, George Kesidis, Christopher M. Collins

https://doi.org/10.1162/neco_a_00284 | pp. 1926-1966

We introduce new inductive, generative semisupervised mixtures with more finely grained class label generation mechanisms than in previous work. Our models combine advantages of semisupervised mixtures, which achieve label extrapolation over a component, and nearest-neighbor (NN)/nearest-prototype (NP) classification, which achieve accurate classification in the vicinity of labeled samples or prototypes. For our NN-based method, we propose a novel two-stage stochastic data generation, with all samples first generated using a standard finite mixture and then all class labels generated, conditioned on the samples and their components of origin. This mechanism entails an underlying Markov random field, specific to each mixture component or cluster. We invoke the pseudo-likelihood formulation, which forms the basis for an approximate generalized expectation-maximization model learning algorithm. Our NP-based model overcomes a problem with the NN-based model that manifests at very low labeled fractions. Both models are advantageous when within-component class proportions are not constant over the feature space region “owned by” a component. The practicality of this scenario is borne out by experiments on UC Irvine data sets, which demonstrate significant gains in classification accuracy over previous semisupervised mixtures and also overall gains, over KNN classification. Moreover, for very small labeled fractions, our methods overall outperform supervised linear and nonlinear kernel support vector machines.

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57. An Efficient Learning Procedure for Deep Boltzmann Machines

Ruslan Salakhutdinov, Geoffrey E. Hinton

https://doi.org/10.1162/neco_a_00311 | pp. 1967-2006

We present a new learning algorithm for Boltzmann machines that contain many layers of hidden variables. Data-dependent statistics are estimated using a variational approximation that tends to focus on a single mode, and data-independent statistics are estimated using persistent Markov chains. The use of two quite

different techniques for estimating the two types of statistic that enter into the gradient of the log likelihood makes it practical to learn Boltzmann machines with multiple hidden layers and millions of parameters. The learning can be made more efficient by using a layer-by-layer pretraining phase that initializes the weights sensibly. The pretraining also allows the variational inference to be initialized sensibly with a single bottom-up pass. We present results on the MNIST and NORB data sets showing that deep Boltzmann machines learn very good generative models of handwritten digits and 3D objects. We also show that the features discovered by deep Boltzmann machines are a very effective way to initialize the hidden layers of feedforward neural nets, which are then discriminatively fine-tuned.

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58. A Framework for Evaluating Pairwise and Multiway Synchrony Among Stimulus-Driven Neurons

Ryan C. Kelly, Robert E. Kass

https://doi.org/10.1162/neco_a_00307 | pp. 2007-2032

Several authors have previously discussed the use of log-linear models, often called maximum entropy models, for analyzing spike train data to detect synchrony. The usual log-linear modeling techniques, however, do not allow time-varying firing rates that typically appear in stimulus-driven (or action-driven) neurons, nor do they incorporate non-Poisson history effects or covariate effects. We generalize the usual approach, combining point-process regression models of individual neuron activity with log-linear models of multiway synchronous interaction. The methods are illustrated with results found in spike trains recorded simultaneously from primary visual cortex. We then assess the amount of data needed to reliably detect multiway spiking.

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59. Competition Through Selective Inhibitory Synchrony

Ueli Rutishauser, Jean-Jacques Slotine, Rodney J. Douglas

https://doi.org/10.1162/neco_a_00304 | pp. 2033-2052

Models of cortical neuronal circuits commonly depend on inhibitory feedback to control gain, provide signal normalization, and selectively amplify signals using winner-take-all (WTA) dynamics. Such models generally assume that excitatory and inhibitory neurons are able to interact easily because their axons and dendrites are colocalized in the same small volume. However, quantitative neuroanatomical studies of the dimensions of axonal and dendritic trees of neurons in the neocortex show that this colocalization assumption is not valid. In this letter, we describe a simple modification to the WTA circuit design that permits the effects of distributed inhibitory neurons to be coupled through synchronization, and so allows a single WTA to be distributed widely in cortical space, well beyond the arborization of any single inhibitory neuron and even across different cortical areas. We prove by nonlinear contraction analysis and demonstrate by simulation that distributed WTA subsystems combined by such inhibitory synchrony are inherently stable. We show analytically that synchronization is substantially faster than winner selection. This circuit mechanism allows networks of independent WTAs to fully or partially compete with other.

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60. Information Recall Using Relative Spike Timing in a Spiking Neural Network

Philip Sterne

https://doi.org/10.1162/neco_a_00306 | pp. 2053-2077

We present a neural network that is capable of completing and correcting a spiking pattern given only a partial, noisy version. It operates in continuous time and represents information using the relative timing of individual spikes. The network is capable of correcting and recalling multiple patterns simultaneously. We analyze the network's performance in terms of information recall. We explore two measures of the capacity of the network: one that values the accurate recall of individual spike times and another that values only the presence or absence of complete patterns. Both measures of information are found to scale linearly in both the number of neurons and the period of the patterns, suggesting these are natural measures of network information. We show a smooth transition from encodings that provide precise spike times to flexible encodings that can encode many scenes. This makes it plausible that many diverse tasks could be learned with such an encoding.

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61. Neural Dynamics, Bifurcations, and Firing Rates in a Quadratic Integrate-and-Fire Model with a Recovery Variable. I: Deterministic Behavior

Eli Shlizerman, Philip Holmes

https://doi.org/10.1162/neco_a_00308 | pp. 2078-2118

We study the dynamics of a quadratic integrate-and-fire model of a single-compartment neuron with a slow recovery variable, as input current and parameters describing timescales, recovery variable, and postspike reset change. Analysis of a codimension 2 bifurcation reveals that the domain of attraction of a stable hyperpolarized rest state interacts subtly with reset parameters, which reposition the system state after spiking. We obtain explicit approximations of instantaneous firing rates for fixed values of the recovery variable, and use the averaging theorem to obtain asymptotic firing rates as a function of current and reset parameters. Along with the different phase-plane geometries, these computations provide explicit tools for the interpretation of different spiking patterns and guide parameter selection in modeling different cortical cell types.

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62. Multinomial Bayesian Learning for Modeling Classical and Nonclassical Receptive Field Properties

Haruo Hosoya

https://doi.org/10.1162/neco_a_00310 | pp. 2119-2150

We study the interplay of Bayesian inference and natural image learning in a hierarchical vision system, in relation to the response properties of early visual cortex. We particularly focus on a Bayesian network with multinomial variables that can represent discrete feature spaces similar to hypercolumns combining minicolumns, enforce sparsity of activation to learn efficient representations, and explain divisive normalization. We demonstrate that maximal-likelihood learning using sampling-based Bayesian inference gives rise to classical receptive field properties similar to V1 simple cells and V2 cells, while inference performed on the trained network yields nonclassical context-dependent response properties such as

cross-orientation suppression and filling in. Comparison with known physiological properties reveals some qualitative and quantitative similarities.

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63. Learning Where to Attend with Deep Architectures for Image Tracking

Misha Denil, Loris Bazzani, Hugo Larochelle, Nando de Freitas

https://doi.org/10.1162/neco_a_00312 | pp. 2151-2184

We discuss an attentional model for simultaneous object tracking and recognition that is driven by gaze data. Motivated by theories of perception, the model consists of two interacting pathways, identity and control, intended to mirror the what and where pathways in neuroscience models. The identity pathway models object appearance and performs classification using deep (factored)-restricted Boltzmann machines. At each point in time, the observations consist of foveated images, with decaying resolution toward the periphery of the gaze. The control pathway models the location, orientation, scale, and speed of the attended object. The posterior distribution of these states is estimated with particle filtering. Deeper in the control pathway, we encounter an attentional mechanism that learns to select gazes so as to minimize tracking uncertainty. Unlike in our previous work, we introduce gaze selection strategies that operate in the presence of partial information and on a continuous action space. We show that a straightforward extension of the existing approach to the partial information setting results in poor performance, and we propose an alternative method based on modeling the reward surface as a gaussian process. This approach gives good performance in the presence of partial information and allows us to expand the action space from a small, discrete set of fixation points to a continuous domain.

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64. Chaotic Exploration and Learning of Locomotion Behaviors

Yoonsik Shim, Phil Husbands

https://doi.org/10.1162/neco_a_00313 | pp. 2185-2222

We present a general and fully dynamic neural system, which exploits intrinsic chaotic dynamics, for the real-time goal-directed exploration and learning of the possible locomotion patterns of an articulated robot of an arbitrary morphology in an unknown environment. The controller is modeled as a network of neural oscillators that are initially coupled only through physical embodiment, and goal-directed exploration of coordinated motor patterns is achieved by chaotic search using adaptive bifurcation. The phase space of the indirectly coupled neural-body-environment system contains multiple transient or permanent self-organized dynamics, each of which is a candidate for a locomotion behavior. The adaptive bifurcation enables the system orbit to wander through various phase-coordinated states, using its intrinsic chaotic dynamics as a driving force, and stabilizes on to one of the states matching the given goal criteria. In order to improve the sustainability of useful transient patterns, sensory homeostasis has been introduced, which results in an increased diversity of motor outputs, thus achieving multiscale exploration. A rhythmic pattern discovered by this process is memorized and sustained by changing the wiring between initially disconnected oscillators using an adaptive synchronization method. Our results show that the novel neurorobotic system is able to create and learn multiple locomotion behaviors for a wide range of body configurations and physical environments and can readapt in realtime after sustaining damage.

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65. Strictly Positive-Definite Spike Train Kernels for Point-Process Divergences

Il Memming Park, Sohan Seth, Murali Rao, José C. Pr■■ncipe

https://doi.org/10.1162/neco_a_00309 | pp. 2223-2250

Exploratory tools that are sensitive to arbitrary statistical variations in spike train observations open up the possibility of novel neuroscientific discoveries. Developing such tools, however, is difficult due to the lack of Euclidean structure of the spike train space, and an experimenter usually prefers simpler tools that capture only limited statistical features of the spike train, such as mean spike count or mean firing rate. We explore strictly positive-definite kernels on the space of spike trains to offer both a structural representation of this space and a platform for developing statistical measures that explore features beyond count or rate. We apply these kernels to construct measures of divergence between two point processes and use them for hypothesis testing, that is, to observe if two sets of spike trains originate from the same underlying probability law. Although there exist positive-definite spike train kernels in the literature, we establish that these kernels are not strictly definite and thus do not induce measures of divergence. We discuss the properties of both of these existing nonstrict kernels and the novel strict kernels in terms of their computational complexity, choice of free parameters, and performance on both synthetic and real data through kernel principal component analysis and hypothesis testing.

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66. Frequency Selectivity Emerging from Spike-Timing-Dependent Plasticity

Matthieu Gilson, Moritz Bürck, Anthony N. Burkitt, J. Leo van Hemmen

https://doi.org/10.1162/neco_a_00331 | pp. 2251-2279

Periodic neuronal activity has been observed in various areas of the brain, from lower sensory to higher cortical levels. Specific frequency components contained in this periodic activity can be identified by a neuronal circuit that behaves as a bandpass filter with given preferred frequency, or best modulation frequency (BMF). For BMFs typically ranging from 10 to 200 Hz, a plausible and minimal configuration consists of a single neuron with adjusted excitatory and inhibitory synaptic connections. The emergence, however, of such a neuronal circuitry is still unclear. In this letter, we demonstrate how spike-timing-dependent plasticity (STDP) can give rise to frequency-dependent learning, thus leading to an input selectivity that enables frequency identification. We use an in-depth mathematical analysis of the learning dynamics in a population of plastic inhibitory connections. These provide inhomogeneous postsynaptic responses that depend on their dendritic location. We find that synaptic delays play a crucial role in organizing the weight specialization induced by STDP. Under suitable conditions on the synaptic delays and postsynaptic potentials (PSPs), the BMF of a neuron after learning can match the training frequency. In particular, proximal (distal) synapses with shorter (longer) dendritic delay and somatically measured PSP time constants respond better to higher (lower) frequencies. As a result, the neuron will respond maximally to any stimulating frequency (in a given range) with which it has been trained in an unsupervised manner. The model predicts that synapses responding to a given BMF form clusters on dendritic branches.

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67. Optimal Population Codes for Space: Grid Cells Outperform Place Cells

Alexander Mathis, Andreas V. M. Herz, Martin Stemmler

https://doi.org/10.1162/neco_a_00319 | pp. 2280-2317

Rodents use two distinct neuronal coordinate systems to estimate their position: place fields in the hippocampus and grid fields in the entorhinal cortex. Whereas place cells spike at only one particular spatial location, grid cells fire at multiple sites that correspond to the points of an imaginary hexagonal lattice. We study how to best construct place and grid codes, taking the probabilistic nature of neural spiking into account. Which spatial encoding properties of individual neurons confer the highest resolution when decoding the animal's position from the neuronal population response? A priori, estimating a spatial position from a grid code could be ambiguous, as regular periodic lattices possess translational symmetry. The solution to this problem requires lattices for grid cells with different spacings; the spatial resolution crucially depends on choosing the right ratios of these spacings across the population. We compute the expected error in estimating the position in both the asymptotic limit, using Fisher information, and for low spike counts, using maximum likelihood estimation. Achieving high spatial resolution and covering a large range of space in a grid code leads to a trade-off: the best grid code for spatial resolution is built of nested modules with different spatial periods, one inside the other, whereas maximizing the spatial range requires distinct spatial periods that are pairwisely incommensurate. Optimizing the spatial resolution predicts two grid cell properties that have been experimentally observed. First, short lattice spacings should outnumber long lattice spacings. Second, the grid code should be self-similar across different lattice spacings, so that the grid field always covers a fixed fraction of the lattice period. If these conditions are satisfied and the spatial "tuning curves" for each neuron span the same range of firing rates, then the resolution of the grid code easily exceeds that of the best possible place code with the same number of neurons.

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68. Distinguishing the Causes of Firing with the Membrane Potential Slope

Achilleas Koutsou, Chris Christodoulou, Guido Bugmann, Jacob Kanev

https://doi.org/10.1162/neco_a_00323 | pp. 2318-2345

In this letter, we aim to measure the relative contribution of coincidence detection and temporal integration to the firing of spikes of a simple neuron model. To this end, we develop a method to infer the degree of synchrony in an ensemble of neurons whose firing drives a single postsynaptic cell. This is accomplished by studying the effects of synchronous inputs on the membrane potential slope of the neuron and estimating the degree of response-relevant input synchrony, which determines the neuron's operational mode. The measure is calculated using the normalized slope of the membrane potential prior to the spikes fired by a neuron, and we demonstrate that it is able to distinguish between the two operational modes. By applying this measure to the membrane potential time course of a leaky integrate-and-fire neuron with the partial somatic reset mechanism, which has been shown to be the most likely candidate to reflect the mechanism used in the brain for reproducing the highly irregular firing at high rates, we show that the partial reset model operates as a temporal integrator of incoming excitatory postsynaptic potentials and that coincidence detection is not necessary for producing such high irregular firing.

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69. Hebbian Learning of Recurrent Connections: A Geometrical Perspective

Mathieu Galtier, Olivier Faugeras, Paul C. Bressloff

https://doi.org/10.1162/neco_a_00322 | pp. 2346-2383

We show how a Hopfield network with modifiable recurrent connections undergoing slow Hebbian learning can extract the underlying geometry of an input space. First, we use a slow and fast analysis to derive an averaged system whose dynamics derives from an energy function and therefore always converges to equilibrium points. The equilibria reflect the correlation structure of the inputs, a global object extracted through local recurrent interactions only. Second, we use numerical methods to illustrate how learning extracts the hidden geometrical structure of the inputs. Indeed, multidimensional scaling methods make it possible to project the final connectivity matrix onto a Euclidean distance matrix in a high-dimensional space, with the neurons labeled by spatial position within this space. The resulting network structure turns out to be roughly convolutional. The residual of the projection defines the nonconvolutional part of the connectivity, which is minimized in the process. Finally, we show how restricting the dimension of the space where the neurons live gives rise to patterns similar to cortical maps. We motivate this using an energy efficiency argument based on wire length minimization. Finally, we show how this approach leads to the emergence of ocular dominance or orientation columns in primary visual cortex via the self-organization of recurrent rather than feedforward connections. In addition, we establish that the nonconvolutional (or long-range) connectivity is patchy and is co-aligned in the case of orientation learning.

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70. Characterizing Responses of Translation-Invariant Neurons to Natural Stimuli: Maximally Informative Invariant Dimensions

Michael Eickenberg, Ryan J. Rowekamp, Minjoon Kouh, Tatyana O. Sharpee

https://doi.org/10.1162/neco_a_00330 | pp. 2384-2421

The human visual system is capable of recognizing complex objects even when their appearances change drastically under various viewing conditions. Especially in the higher cortical areas, the sensory neurons reflect such functional capacity in their selectivity for complex visual features and invariance to certain object transformations, such as image translation. Due to the strong nonlinearities necessary to achieve both the selectivity and invariance, characterizing and predicting the response patterns of these neurons represents a formidable computational challenge. A related problem is that such neurons are poorly driven by randomized inputs, such as white noise, and respond strongly only to stimuli with complex high-order correlations, such as natural stimuli. Here we describe a novel two-step optimization technique that can characterize both the shape selectivity and the range and coarseness of position invariance from neural responses to natural stimuli. One step in the optimization is finding the template as the maximally informative dimension given the estimated spatial location where the response could have been triggered within each image. The estimates of the locations that triggered the response are updated in the next step. Under the assumption of a monotonic relationship between the firing rate and stimulus projections on the template at a given position, the most likely location is the one that has the largest projection on the estimate of the template. The algorithm shows quick convergence during optimization, and the estimation results are reliable even in the regime of small signal-to-noise ratios. When we apply the algorithm to responses of complex cells in the primary visual cortex (V1) to natural movies, we find that responses of the majority of cells were significantly better described by translation-invariant models based on one template compared with position-specific models with several relevant features.

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71. Sparse Coding on the Spot: Spontaneous Retinal Waves Suffice for Orientation Selectivity

Jonathan J. Hunt, Michael R. Ibbotson, Geoffrey J. Goodhill

https://doi.org/10.1162/neco_a_00333 | pp. 2422-2433

Ohshiro, Hussain, and Weliky (2011) recently showed that ferrets reared with exposure to flickering spot stimuli, in the absence of oriented visual experience, develop oriented receptive fields. They interpreted this as refutation of efficient coding models, which require oriented input in order to develop oriented receptive fields. Here we show that these data are compatible with the efficient coding hypothesis if the influence of spontaneous retinal waves is considered. We demonstrate that independent component analysis learns predominantly oriented receptive fields when trained on a mixture of spot stimuli and spontaneous retinal waves. Further, we show that the efficient coding hypothesis provides a compelling explanation for the contrast between the lack of receptive field changes seen in animals reared with spot stimuli and the significant cortical reorganisation observed in stripe-reared animals.

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72. Bayesian Community Detection

Morten Mørup, Mikkel N. Schmidt

https://doi.org/10.1162/neco_a_00314 | pp. 2434-2456

Many networks of scientific interest naturally decompose into clusters or communities with comparatively fewer external than internal links; however, current Bayesian models of network communities do not exert this intuitive notion of communities. We formulate a nonparametric Bayesian model for community detection consistent with an intuitive definition of communities and present a Markov chain Monte Carlo procedure for inferring the community structure. A Matlab toolbox with the proposed inference procedure is available for download. On synthetic and real networks, our model detects communities consistent with ground truth, and on real networks, it outperforms existing approaches in predicting missing links. This suggests that community structure is an important structural property of networks that should be explicitly modeled.

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73. Learning the Dynamics of Objects by Optimal Functional Interpolation

Jong-Hoon Ahn, In Young Kim

https://doi.org/10.1162/neco_a_00325 | pp. 2457-2472

Many areas of science and engineering rely on functional data and their numerical analysis. The need to analyze time-varying functional data raises the general problem of interpolation, that is, how to learn a smooth time evolution from a finite number of observations. Here, we introduce optimal functional interpolation (OFI), a numerical algorithm that interpolates functional data over time. Unlike the usual interpolation or learning algorithms, the OFI algorithm obeys the continuity equation, which describes the transport of some types of conserved quantities, and its implementation shows smooth, continuous flows of quantities. Without the need to take into account equations of motion such as the Navier-Stokes equation or the diffusion equation, OFI is capable of learning the dynamics of objects such as those represented by mass, image intensity, particle concentration, heat, spectral density, and probability density.

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74. Inhibition in Multiclass Classification

Ramón Huerta, Shankar Vembu, José M. Amigó, Thomas Nowotny, Charles Elkan

https://doi.org/10.1162/neco_a_00321 | pp. 2473-2507

The role of inhibition is investigated in a multiclass support vector machine formalism inspired by the brain structure of insects. The so-called mushroom bodies have a set of output neurons, or classification functions, that compete with each other to encode a particular input. Strongly active output neurons depress or inhibit the remaining outputs without knowing which is correct or incorrect. Accordingly, we propose to use a classification function that embodies unselective inhibition and train it in the large margin classifier framework. Inhibition leads to more robust classifiers in the sense that they perform better on larger areas of appropriate hyperparameters when assessed with leave-one-out strategies. We also show that the classifier with inhibition is a tight bound to probabilistic exponential models and is Bayes consistent for 3-class problems. These properties make this approach useful for data sets with a limited number of labeled examples. For larger data sets, there is no significant comparative advantage to other multiclass SVM approaches.

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75. Multilabel Classification with Principal Label Space Transformation

Farbound Tai, Hsuan-Tien Lin

https://doi.org/10.1162/neco_a_00320 | pp. 2508-2542

We consider a hypercube view to perceive the label space of multilabel classification problems geometrically. The view allows us not only to unify many existing multilabel classification approaches but also design a novel algorithm, principal label space transformation (PLST), that captures key correlations between labels before learning. The simple and efficient PLST relies on only singular value decomposition as the key step. We derive the theoretical guarantee of PLST and evaluate its empirical performance using real-world data sets. Experimental results demonstrate that PLST is faster than the traditional binary relevance approach and is superior to the modern compressive sensing approach in terms of both accuracy and efficiency.

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76. Mapping of Visual Receptive Fields by Tomographic Reconstruction

Gordon Pipa, Zhe Chen, Sergio Neuenschwander, Bruss Lima, Emery N. Brown

https://doi.org/10.1162/neco_a_00334 | pp. 2543-2578

The moving bar experiment is a classic paradigm for characterizing the receptive field (RF) properties of neurons in primary visual cortex (V1). Current approaches for analyzing neural spiking activity recorded from these experiments do not take into account the point-process nature of these data and the circular geometry of the stimulus presentation. We present a novel analysis approach to mapping V1 receptive fields that combines point-process generalized linear models (PPGLM) with tomographic reconstruction computed by filtered-back projection. We use the method to map the RF sizes and orientations of 251 V1 neurons recorded from two macaque monkeys during a moving bar experiment. Our cross-validated goodness-of-fit analyses show that the PPGLM provides a more accurate characterization of spike train data than analyses based on rate functions computed by the methods of spike-triggered averages or first-order Wiener-Volterra kernel. Our analysis leads to a new definition of RF size as the spatial area over which the spiking activity is

significantly greater than baseline activity. Our approach yields larger RF sizes and sharper orientation tuning estimates. The tomographic reconstruction paradigm further suggests an efficient approach to choosing the number of directions and the number of trials per direction in designing moving bar experiments. Our results demonstrate that standard tomographic principles for image reconstruction can be adapted to characterize V1 RFs and that two fundamental properties, size and orientation, may be substantially different from what is currently reported.

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77. Unsupervised Formation of Vocalization-Sensitive Neurons: A Cortical Model Based on Short-Term and Homeostatic Plasticity

Tyler P. Lee, Dean V. Buonomano

https://doi.org/10.1162/neco_a_00345 | pp. 2579-2603

The discrimination of complex auditory stimuli relies on the spatiotemporal structure of spike patterns arriving in the cortex. While recordings from auditory areas reveal that many neurons are highly selective to specific spatiotemporal stimuli, the mechanisms underlying this selectivity are unknown. Using computer simulations, we show that selectivity can emerge in neurons in an entirely unsupervised manner. The model is based on recurrently connected spiking neurons and synapses that exhibit short-term synaptic plasticity. During a developmental stage, spoken digits were presented to the network; the only type of long-term plasticity present was a form of homeostatic synaptic plasticity. From an initially unresponsive state, training generated a high percentage of neurons that responded selectively to individual digits. Furthermore, units within the network exhibited a cardinal feature of vocalization-sensitive neurons in vivo: differential responses between forward and reverse stimulus presentations. Direction selectivity deteriorated significantly, however, if short-term synaptic plasticity was removed. These results establish that a simple form of homeostatic plasticity is capable of guiding recurrent networks into regimes in which complex stimuli can be discriminated. In addition, one computational function of short-term synaptic plasticity may be to provide an inherent temporal asymmetry, thus contributing to the characteristic forward-reverse selectivity.

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78. The Rise and Fall of Memory in a Model of Synaptic Integration

Terry Elliott, Konstantinos Lagogiannis

https://doi.org/10.1162/neco_a_00335 | pp. 2604-2654

Plasticity-inducing stimuli must typically be presented many times before synaptic plasticity is expressed, perhaps because induction signals gradually accumulate before overt strength changes occur. We consider memory dynamics in a mathematical model with synapses that integrate plasticity induction signals before expressing plasticity. We find that the memory trace initially rises before reaching a maximum and then falling. The memory signal dissociates into separate oblivescence and reminiscence components, with reminiscence initially dominating recall. In radical contrast, related but nonintegrative models exhibit only a highly problematic oblivescence. Synaptic integration mechanisms possess natural timescales, depending on the statistics of the induction signals. Together with neuromodulation, these timescales may therefore also begin to provide a natural account of the well-known spacing effect in the transition to late-phase plasticity. Finally, we propose experiments that could distinguish between integrative and nonintegrative synapses. Such experiments should further elucidate the synaptic signal processing mechanisms postulated by our model.

79. Predicting Single-Neuron Activity in Locally Connected Networks

Feraz Azhar, William S. Anderson

https://doi.org/10.1162/neco_a_00343 | pp. 2655-2677

The characterization of coordinated activity in neuronal populations has received renewed interest in the light of advancing experimental techniques that allow recordings from multiple units simultaneously. Across both in vitro and in vivo preparations, nearby neurons show coordinated responses when spontaneously active and when subject to external stimuli. Recent work (Truccolo, Hochberg, & Donoghue, 2010) has connected these coordinated responses to behavior, showing that small ensembles of neurons in arm-related areas of sensorimotor cortex can reliably predict single-neuron spikes in behaving monkeys and humans. We investigate this phenomenon using an analogous point process model, showing that in the case of a computational model of cortex responding to random background inputs, one is similarly able to predict the future state of a single neuron by considering its own spiking history, together with the spiking histories of randomly sampled ensembles of nearby neurons. This model exhibits realistic cortical architecture and displays bursting episodes in the two distinct connectivity schemes studied. We conjecture that the baseline predictability we find in these instances is characteristic of locally connected networks more broadly considered.

80. Nearly Extensive Sequential Memory Lifetime Achieved by Coupled Nonlinear Neurons

Taro Toyoizumi

https://doi.org/10.1162/neco_a_00324 | pp. 2678-2699

Many cognitive processes rely on the ability of the brain to hold sequences of events in short-term memory. Recent studies have revealed that such memory can be read out from the transient dynamics of a network of neurons. However, the memory performance of such a network in buffering past information has been rigorously estimated only in networks of linear neurons. When signal gain is kept low, so that neurons operate primarily in the linear part of their response nonlinearity, the memory lifetime is bounded by the square root of the network size. In this work, I demonstrate that it is possible to achieve a memory lifetime almost proportional to the network size, "an extensive memory lifetime," when the nonlinearity of neurons is appropriately used. The analysis of neural activity revealed that nonlinear dynamics prevented the accumulation of noise by partially removing noise in each time step. With this error-correcting mechanism, I demonstrate that a memory lifetime of order N/logN can be achieved.

81. Replicating Receptive Fields of Simple and Complex Cells in Primary Visual Cortex in a Neuronal Network Model with Temporal and Population Sparseness and Reliability

Takuma Tanaka, Toshio Aoyagi, Takeshi Kaneko

https://doi.org/10.1162/neco_a_00341 | pp. 2700-2725

We propose a new principle for replicating receptive field properties of neurons in the primary visual cortex. We derive a learning rule for a feedforward network, which maintains a low firing rate for the output neurons

(resulting in temporal sparseness) and allows only a small subset of the neurons in the network to fire at any given time (resulting in population sparseness). Our learning rule also sets the firing rates of the output neurons at each time step to near-maximum or near-minimum levels, resulting in neuronal reliability. The learning rule is simple enough to be written in spatially and temporally local forms. After the learning stage is performed using input image patches of natural scenes, output neurons in the model network are found to exhibit simple-cell-like receptive field properties. When the output of these simple-cell-like neurons are input to another model layer using the same learning rule, the second-layer output neurons after learning become less sensitive to the phase of gratings than the simple-cell-like input neurons. In particular, some of the second-layer output neurons become completely phase invariant, owing to the convergence of the connections from first-layer neurons with similar orientation selectivity to second-layer neurons in the model network. We examine the parameter dependencies of the receptive field properties of the model neurons after learning and discuss their biological implications. We also show that the localized learning rule is consistent with experimental results concerning neuronal plasticity and can replicate the receptive fields of simple and complex cells.

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82. Motion-Based Prediction Is Sufficient to Solve the Aperture Problem

Laurent U. Perrinet, Guillaume S. Masson

https://doi.org/10.1162/neco_a_00332 | pp. 2726-2750

In low-level sensory systems, it is still unclear how the noisy information collected locally by neurons may give rise to a coherent global percept. This is well demonstrated for the detection of motion in the aperture problem: as luminance of an elongated line is symmetrical along its axis, tangential velocity is ambiguous when measured locally. Here, we develop the hypothesis that motion-based predictive coding is sufficient to infer global motion. Our implementation is based on a context-dependent diffusion of a probabilistic representation of motion. We observe in simulations a progressive solution to the aperture problem similar to physiology and behavior. We demonstrate that this solution is the result of two underlying mechanisms. First, we demonstrate the formation of a tracking behavior favoring temporally coherent features independent of their texture. Second, we observe that incoherent features are explained away, while coherent information diffuses progressively to the global scale. Most previous models included ad hoc mechanisms such as end-stopped cells or a selection layer to track specific luminance-based features as necessary conditions to solve the aperture problem. Here, we have proved that motion-based predictive coding, as it is implemented in this functional model, is sufficient to solve the aperture problem. This solution may give insights into the role of prediction underlying a large class of sensory computations.

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83. Nonlinearities and Adaptation of Color Vision from Sequential Principal Curves Analysis

Valero Laparra, Sandra Jiménez, Gustavo Camps-Valls, Jesús Malo

https://doi.org/10.1162/neco_a_00342 | pp. 2751-2788

Mechanisms of human color vision are characterized by two phenomenological aspects: the system is nonlinear and adaptive to changing environments. Conventional attempts to derive these features from statistics use separate arguments for each aspect. The few statistical explanations that do consider both phenomena simultaneously follow parametric formulations based on empirical models. Therefore, it may be

argued that the behavior does not come directly from the color statistics but from the convenient functional form adopted. In addition, many times the whole statistical analysis is based on simplified databases that disregard relevant physical effects in the input signal, as, for instance, by assuming flat Lambertian surfaces. In this work, we address the simultaneous statistical explanation of the nonlinear behavior of achromatic and chromatic mechanisms in a fixed adaptation state and the change of such behavior (i.e., adaptation) under the change of observation conditions. Both phenomena emerge directly from the samples through a single data-driven method: the sequential principal curves analysis (SPCA) with local metric. SPCA is a new manifold learning technique to derive a set of sensors adapted to the manifold using different optimality criteria. Here sequential refers to the fact that sensors (curvilinear dimensions) are designed one after the other, and not to the particular (eventually iterative) method to draw a single principal curve. Moreover, in order to reproduce the empirical adaptation reported under D65 and A illuminations, a new database of colorimetrically calibrated images of natural objects under these illuminants was gathered, thus overcoming the limitations of available databases. The results obtained by applying SPCA show that the psychophysical behavior on color discrimination thresholds, discount of the illuminant, and corresponding pairs in asymmetric color matching emerge directly from realistic data regularities, assuming no a priori functional form. These results provide stronger evidence for the hypothesis of a statistically driven organization of color sensors. Moreover, the obtained results suggest that the nonuniform resolution of color sensors at this low abstraction level may be guided by an error-minimization strategy rather than by an information-maximization goal.

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84. An Extension of the Receiver Operating Characteristic Curve and AUC-Optimal Classification

Takashi Takenouchi, Osamu Komori, Shinto Eguchi

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While most proposed methods for solving classification problems focus on minimization of the classification error rate, we are interested in the receiver operating characteristic (ROC) curve, which provides more information about classification performance than the error rate does. The area under the ROC curve (AUC) is a natural measure for overall assessment of a classifier based on the ROC curve. We discuss a class of concave functions for AUC maximization in which a boosting-type algorithm including RankBoost is considered, and the Bayesian risk consistency and the lower bound of the optimum function are discussed. A procedure derived by maximizing a specific optimum function has high robustness, based on gross error sensitivity. Additionally, we focus on the partial AUC, which is the partial area under the ROC curve. For example, in medical screening, a high true-positive rate to the fixed lower false-positive rate is preferable and thus the partial AUC corresponding to lower false-positive rates is much more important than the remaining AUC. We extend the class of concave optimum functions for partial AUC optimality with the boosting algorithm. We investigated the validity of the proposed method through several experiments with data sets in the UCI repository.

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85. Adaptive Metric Learning Vector Quantization for Ordinal Classification

Shereen Fouad, Peter Ti■o

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Many pattern analysis problems require classification of examples into naturally ordered classes. In such cases, nominal classification schemes will ignore the class order relationships, which can have a detrimental effect on classification accuracy. This article introduces two novel ordinal learning vector quantization (LVQ) schemes, with metric learning, specifically designed for classifying data items into ordered classes. In ordinal LVQ, unlike in nominal LVQ, the class order information is used during training in selecting the class prototypes to be adapted, as well as in determining the exact manner in which the prototypes get updated. Prototype-based models in general are more amenable to interpretations and can often be constructed at a smaller computational cost than alternative nonlinear classification models. Experiments demonstrate that the proposed ordinal LVQ formulations compare favorably with their nominal counterparts. Moreover, our methods achieve competitive performance against existing benchmark ordinal regression models.

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86. A Network of Spiking Neurons for Computing Sparse Representations in an Energy-Efficient Way

Tao Hu, Alexander Genkin, Dmitri B. Chklovskii

https://doi.org/10.1162/neco_a_00353 | pp. 2852-2872

Computing sparse redundant representations is an important problem in both applied mathematics and neuroscience. In many applications, this problem must be solved in an energy-efficient way. Here, we propose a hybrid distributed algorithm (HDA), which solves this problem on a network of simple nodes communicating by low-bandwidth channels. HDA nodes perform both gradient-descent-like steps on analog internal variables and coordinate-descent-like steps via quantized external variables communicated to each other. Interestingly, the operation is equivalent to a network of integrate-and-fire neurons, suggesting that HDA may serve as a model of neural computation. We show that the numerical performance of HDA is on par with existing algorithms. In the asymptotic regime, the representation error of HDA decays with time, t, as 1/t. HDA is stable against time-varying noise; specifically, the representation error decays as $1/\sqrt{t}$ for gaussian white noise.

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87. The Hippocampus as a Stable Memory Allocator for Cortex

Leslie G. Valiant

https://doi.org/10.1162/neco_a_00357 | pp. 2873-2899

It is suggested here that mammalian hippocampus serves as an allocator of neurons in cortex for memorizing new items. A construction of a shallow feedforward network with biologically plausible parameters is given that possesses the characteristics needed for such an allocator. In particular, the construction is stabilizing in that for inputs within a range of activity levels spanning more than an order of magnitude, the output will have activity levels differing as little as 1%. It is also noise tolerant in that pairs of input patterns that differ little will generate output patterns that differ little. Further, pairs of inputs that differ by much will be mapped to outputs that also differ sufficiently that they can be treated by cortex as distinct.

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88. Adaptive Classification on Brain-Computer Interfaces Using Reinforcement Signals

Alberto Llera, Vicenç Gómez, Hilbert J. Kappen
https://doi.org/10.1162/neco_a_00348 | pp. 2900-2923

We introduce a probabilistic model that combines a classifier with an extra reinforcement signal (RS) encoding the probability of an erroneous feedback being delivered by the classifier. This representation computes the class probabilities given the task related features and the reinforcement signal. Using expectation maximization (EM) to estimate the parameter values under such a model shows that some existing adaptive classifiers are particular cases of such an EM algorithm. Further, we present a new algorithm for adaptive classification, which we call constrained means adaptive classifier, and show using EEG data and simulated RS that this classifier is able to significantly outperform state-of-the-art adaptive classifiers.

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89. The Basal Ganglia Optimize Decision Making over General Perceptual Hypotheses

Nathan F. Lepora, Kevin Gurney
https://doi.org/10.1162/neco_a_00360 | pp. 2924-2945

The basal ganglia are a subcortical group of interconnected nuclei involved in mediating action selection within cortex. A recent proposal is that this selection leads to optimal decision making over multiple alternatives because the basal ganglia anatomy maps onto a network implementation of an optimal statistical method for hypothesis testing, assuming that cortical activity encodes evidence for constrained gaussian-distributed alternatives. This letter demonstrates that this model of the basal ganglia extends naturally to encompass general Bayesian sequential analysis over arbitrary probability distributions, which raises the proposal to a practically realizable theory over generic perceptual hypotheses. We also show that the evidence in this model can represent either log likelihoods, log-likelihood ratios, or log odds, all leading proposals for the cortical processing of sensory data. For these reasons, we claim that the basal ganglia optimize decision making over general perceptual hypotheses represented in cortex. The relation of this theory to cortical encoding, cortico-basal ganglia anatomy, and reinforcement learning is discussed.

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90. A Neural Circuit for Robust Time-to-Contact Estimation Based on Primate MST

N. Andrew Browning
https://doi.org/10.1162/neco_a_00347 | pp. 2946-2963

Time-to-contact (TTC) estimation is beneficial for visual navigation. It can be estimated from an image projection, either in a camera or on the retina, by looking at the rate of expansion of an object. When expansion rate (E) is properly defined, TTC = 1/E. Primate dorsal MST cells have receptive field structures suited to the estimation of expansion and TTC. However, the role of MST cells in TTC estimation has been discounted because of large receptive fields, the fact that neither they nor preceding brain areas appear to decompose the motion field to estimate divergence, and a lack of experimental data. This letter demonstrates mathematically that template models of dorsal MST cells can be constructed such that the output of the template match provides an accurate and robust estimate of TTC. The template match extracts the relevant components of the motion field and scales them such that the output of each component of the

template match is an estimate of expansion. It then combines these component estimates to provide a mean estimate of expansion across the object. The output of model MST provides a direct measure of TTC. The ViSTARS model of primate visual navigation was updated to incorporate the modified templates. In ViSTARS and in primates, speed is represented as a population code in V1 and MT. A population code for speed complicates TTC estimation from a template match. Results presented in this letter demonstrate that the updated template model of MST accurately codes TTC across a population of model MST cells. We conclude that the updated template model of dorsal MST simultaneously and accurately codes TTC and heading regardless of receptive field size, object size, or motion representation. It is possible that a subpopulation of MST cells in primates represents expansion in this way.

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91. Regulation of Ambient GABA Levels by Neuron-Glia Signaling for Reliable Perception of Multisensory Events

Osamu Hoshino

https://doi.org/10.1162/neco_a_00356 | pp. 2964-2993

Activities of sensory-specific cortices are known to be suppressed when presented with a different sensory modality stimulus. This is referred to as cross-modal inhibition, for which the conventional synaptic mechanism is unlikely to work. Interestingly, the cross-modal inhibition could be eliminated when presented with multisensory stimuli arising from the same event. To elucidate the underlying neuronal mechanism of cross-modal inhibition and understand its significance for multisensory information processing, we simulated a neural network model. Principal cell to and GABAergic interneuron to glial cell projections were assumed between and within lower-order unimodal networks (X and Y), respectively. Cross-modality stimulation of Y network activated its principal cells, which then depolarized glial cells of X network. This let transporters on the glial cells export GABA molecules into the extracellular space and increased a level of ambient (extrasynaptic) GABA. The ambient GABA molecules were accepted by extrasynaptic GABA(a) receptors and tonically inhibited principal cells of the X network. Cross-modal inhibition took place in a nonsynaptic manner. Identical modality stimulation of X network activated its principal cells, which then activated interneurons and hyperpolarized glial cells of the X network. This let their transporters import (remove) GABA molecules from the extracellular space and reduced tonic inhibitory current in principal cells, thereby improving their gain function. Top-down signals from a higher-order multimodal network (M) contributed to elimination of the cross-modal inhibition when presented with multisensory stimuli that arose from the same event. Tuning into the multisensory event deteriorated if the cross-modal inhibitory mechanism did not work. We suggest that neuron-glia signaling may regulate local ambient GABA levels in order to coordinate cross-modal inhibition and improve neuronal gain function, thereby achieving reliable perception of multisensory events.

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92. Incremental Slow Feature Analysis: Adaptive Low-Complexity Slow Feature Updating from High-Dimensional Input Streams

Varun Raj Kompella, Matthew Luciw, Jürgen Schmidhuber

https://doi.org/10.1162/neco_a_00344 | pp. 2994-3024

We introduce here an incremental version of slow feature analysis (IncSFA), combining candid covariance-free incremental principal components analysis (CCIPCA) and covariance-free incremental

minor components analysis (CIMCA). IncSFA's feature updating complexity is linear with respect to the input dimensionality, while batch SFA's (BSFA) updating complexity is cubic. IncSFA does not need to store, or even compute, any covariance matrices. The drawback to IncSFA is data efficiency: it does not use each data point as effectively as BSFA. But IncSFA allows SFA to be tractably applied, with just a few parameters, directly on high-dimensional input streams (e.g., visual input of an autonomous agent), while BSFA has to resort to hierarchical receptive-field-based architectures when the input dimension is too high. Further, IncSFA's updates have simple Hebbian and anti-Hebbian forms, extending the biological plausibility of SFA. Experimental results show IncSFA learns the same set of features as BSFA and can handle a few cases where BSFA fails.

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93. On Convergence Rates of Mixtures of Polynomial Experts

Eduardo Mendes, Wenxin Jiang

https://doi.org/10.1162/neco_a_00354 | pp. 3025-3051

In this letter, we consider a mixture-of-experts structure where m experts are mixed, with each expert being related to a polynomial regression model of order k. We study the convergence rate of the maximum likelihood estimator in terms of how fast the Hellinger distance of the estimated density converges to the true density, when the sample size n increases. The convergence rate is found to be dependent on both m and k, while certain choices of m and k are found to produce near-optimal convergence rates.

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94. Shared and Specific Independent Components Analysis for Between-Group Comparison

Shahabeddin Vahdat, Mona Maneshi, Christophe Grova, Jean Gotman, Theodore E. Milner

https://doi.org/10.1162/neco_a_00355 | pp. 3052-3090

Independent component analysis (ICA) has been extensively used in individual and within-group data sets in real-world applications, but how can it be employed in a between-groups or conditions design? Here, we propose a new method to embed group membership information into the FastICA algorithm so as to extract components that are either shared between groups or specific to one or a subset of groups. The proposed algorithm is designed to automatically extract the pattern of differences between different experimental groups or conditions. A new constraint is added to the FastICA algorithm to simultaneously deal with the data of multiple groups in a single ICA run. This cost function restricts the specific components of one group to be orthogonal to the subspace spanned by the data of the other groups. As a result of performing a single ICA on the aggregate data of several experimental groups, the entire variability of data sets is used to extract the shared components. The results of simulations show that the proposed algorithm performs better than the regular method in both the reconstruction of the source signals and classification of shared and specific components. Also, the sensitivity to detect variations in the amplitude of shared components across groups is enhanced. A rigorous proof of convergence is provided for the proposed iterative algorithm. Thus, this algorithm is guaranteed to extract and classify shared and specific independent components across different experimental groups and conditions in a systematic way.

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95. Neural Relax

Elisa Benedetti, Marco Budinich

https://doi.org/10.1162/neco_a_00359 | pp. 3091-3110

We propose a new self-organizing algorithm for a feedforward network inspired to an electrostatic problem that turns out to have intimate relations with information maximization.

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96. The Shape of Phase-Resetting Curves in Oscillators with a Saddle Node on an Invariant Circle Bifurcation

Bard Ermentrout, Leon Glass, Bart E. Oldeman

https://doi.org/10.1162/neco_a_00370 | pp. 3111-3125

We introduce a simple two-dimensional model that extends the Poincaré oscillator so that the attracting limit cycle undergoes a saddle node bifurcation on an invariant circle (SNIC) for certain parameter values. Arbitrarily close to this bifurcation, the phase-resetting curve (PRC) continuously depends on parameters, where its shape can be not only primarily positive or primarily negative but also nearly sinusoidal. This example system shows that one must be careful inferring anything about the bifurcation structure of the oscillator from the shape of its PRC.

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97. A Compartmental Model of Linear Resonance and Signal Transfer in Dendrites

Alan Schoen, Ali Salehiomran, Matthew E. Larkum, Erik P. Cook

https://doi.org/10.1162/neco_a_00366 | pp. 3126-3144

Dendrites carry signals between synapses and the soma and play a central role in neural computation. Although they contain many nonlinear ion channels, their signal-transfer properties are linear under some experimental conditions. In experiments with continuous-time inputs, a resonant linear two-port model has been shown to provide a near-perfect fit to the dendrite-to-soma input-output relationship. In this study, we focused on this linear aspect of signal transfer using impedance functions that replace biophysical channel models in order to describe the electrical properties of the dendritic membrane. The membrane impedance model of dendrites preserves the accuracy of the two-port model with minimal computational complexity. Using this approach, we demonstrate two membrane impedance profiles of dendrites that reproduced the experimentally observed two-port results. These impedance profiles demonstrate that the two-port results are compatible with different computational schemes. In addition, our model highlights how dendritic resonance can minimize the location-dependent attenuation of signals at the resonant frequency. Thus, in this model, dendrites function as linear-resonant filters that carry signals between nonlinear computational units.

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98. Exact Event-Driven Implementation for Recurrent Networks of Stochastic Perfect Integrate-and-Fire Neurons

Thibaud Taillefumier, Jonathan Touboul, Marcelo O. Magnasco

https://doi.org/10.1162/neco_a_00346 | pp. 3145-3180

In vivo cortical recording reveals that indirectly driven neural assemblies can produce reliable and temporally precise spiking patterns in response to stereotyped stimulation. This suggests that despite being fundamentally noisy, the collective activity of neurons conveys information through temporal coding. Stochastic integrate-and-fire models delineate a natural theoretical framework to study the interplay of intrinsic neural noise and spike timing precision. However, there are inherent difficulties in simulating their networks' dynamics in silico with standard numerical discretization schemes. Indeed, the well-posedness of the evolution of such networks requires temporally ordering every neuronal interaction, whereas the order of interactions is highly sensitive to the random variability of spiking times. Here, we answer these issues for perfect stochastic integrate-and-fire neurons by designing an exact event-driven algorithm for the simulation of recurrent networks, with delayed Dirac-like interactions. In addition to being exact from the mathematical standpoint, our proposed method is highly efficient numerically. We envision that our algorithm is especially indicated for studying the emergence of polychronized motifs in networks evolving under spike-timing-dependent plasticity with intrinsic noise.

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99. Tuning Low-Voltage-Activated A-Current for Silent Gain Modulation

Ameera X. Patel, Naomi Murphy, Denis Burdakov

https://doi.org/10.1162/neco_a_00373 | pp. 3181-3190

Modulation of stimulus-response gain and stability of spontaneous (unstimulated) firing are both important for neural computation. However, biologically plausible mechanisms that allow these distinct functional capabilities to coexist in the same neuron are poorly defined. Low-threshold, inactivating (A-type) K(+) currents (I(A)) are found in many biological neurons and are historically known for enabling low-frequency firing. By performing simulations using a conductance-based model neuron, here we show that biologically plausible shifts in I(A) conductance and inactivation kinetics produce dissociated effects on gain and intrinsic firing. This enables I(A) to regulate gain without major changes in intrinsic firing rate. Tuning I(A) properties may thus represent a previously unsuspected single-current mechanism of silent gain control in neurons.

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100. Self-Consistent Learning of the Environment

Kukjin Kang, ■■■-■■■ ■■■■■

https://doi.org/10.1162/neco_a_00371 | pp. 3191-3212

We study the Bayesian process to estimate the features of the environment. We focus on two aspects of the Bayesian process: how estimation error depends on the prior distribution of features and how the prior distribution can be learned from experience. The accuracy of the perception is underestimated when each feature of the environment is considered independently because many different features of the environment are usually highly correlated and the estimation error greatly depends on the correlations. The self-consistent learning process renews the prior distribution of correlated features jointly with the estimation of the environment. Here, maximum a posteriori probability (MAP) estimation decreases the effective dimensions of the feature vector. There are critical noise levels in self-consistent learning with MAP estimation, that cause hysteresis behaviors in learning. The self-consistent learning process with stochastic Bayesian estimation (SBE) makes the presumed distribution of environmental features converge to the true distribution for any level of channel noise. However, SBE is less accurate than MAP estimation. We also

discuss another stochastic method of estimation, SBE2, which has a smaller estimation error than SBE without hysteresis.

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101. Information-Geometric Measures for Estimation of Connection Weight Under Correlated Inputs

Yimin Nie, Masami Tatsuno

https://doi.org/10.1162/neco_a_00367 | pp. 3213-3245

The brain processes information in a highly parallel manner. Determination of the relationship between neural spikes and synaptic connections plays a key role in the analysis of electrophysiological data. Information geometry (IG) has been proposed as a powerful analysis tool for multiple spike data, providing useful insights into the statistical interactions within a population of neurons. Previous work has demonstrated that IG measures can be used to infer the connection weight between two neurons in a neural network. This property is useful in neuroscience because it provides a way to estimate learning-induced changes in synaptic strengths from extracellular neuronal recordings. A previous study has shown, however, that this property would hold only when inputs to neurons are not correlated. Since neurons in the brain often receive common inputs, this would hinder the application of the IG method to real data. We investigated the two-neuron-IG measures in higher-order log-linear models to overcome this limitation. First, we mathematically showed that the estimation of uniformly connected synaptic weight can be improved by taking into account higher-order log-linear models. Second, we numerically showed that the estimation can be improved for more general asymmetrically connected networks. Considering the estimated number of the synaptic connections in the brain, we showed that the two-neuron IG measure calculated by the fourth- or fifth-order log-linear model would provide an accurate estimation of connection strength within approximately a 10% error. These studies suggest that the two-neuron IG measure with higher-order log-linear expansion is a robust estimator of connection weight even under correlated inputs, providing a useful analytical tool for real multineuronal spike data.

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102. Design Strategies for Weight Matrices of Echo State Networks

Tobias Strauß, Welf Wustlich, Roger Labahn

https://doi.org/10.1162/neco_a_00374 | pp. 3246-3276

This article develops approaches to generate dynamical reservoirs of echo state networks with desired properties reducing the amount of randomness. It is possible to create weight matrices with a predefined singular value spectrum. The procedure guarantees stability (echo state property). We prove the minimization of the impact of noise on the training process. The resulting reservoir types are strongly related to reservoirs already known in the literature. Our experiments show that well-chosen input weights can improve performance.

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103. Tangent Bundle Curve Completion with Locally Connected Parallel Networks

Guy Ben-Yosef, Ohad Ben■Shahar

https://doi.org/10.1162/neco_a_00365 | pp. 3277-3316

We propose a theory for cortical representation and computation of visually completed curves that are generated by the visual system to fill in missing visual information (e.g., due to occlusions). Recent computational theories and physiological evidence suggest that although such curves do not correspond to explicit image evidence along their length, their construction emerges from corresponding activation patterns of orientation-selective cells in the primary visual cortex. Previous theoretical work modeled these patterns as least energetic 3D curves in the mathematical continuous space R2 × S1, which abstracts the mammalian striate cortex. Here we discuss the biological plausibility of this theory and present a neural architecture that implements it with locally connected parallel networks. Part of this contribution is also a first attempt to bridge the physiological literature on curve completion with the shape problem and a shape theory. We present completion simulations of our model in natural and synthetic scenes and discuss various observations and predictions that emerge from this theory in the context of curve completion.

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104. A Common Network Architecture Efficiently Implements a Variety of Sparsity-Based Inference Problems

Adam S. Charles, Pierre Garrigues, Christopher J. Rozell

https://doi.org/10.1162/neco_a_00372 | pp. 3317-3339

The sparse coding hypothesis has generated significant interest in the computational and theoretical neuroscience communities, but there remain open questions about the exact quantitative form of the sparsity penalty and the implementation of such a coding rule in neurally plausible architectures. The main contribution of this work is to show that a wide variety of sparsity-based probabilistic inference problems proposed in the signal processing and statistics literatures can be implemented exactly in the common network architecture known as the locally competitive algorithm (LCA). Among the cost functions we examine are approximate l(p) norms ($0 \leq p \leq 2$), modified l(p)-norms, block-l1 norms, and reweighted algorithms. Of particular interest is that we show significantly increased performance in reweighted l1 algorithms by inferring all parameters jointly in a dynamical system rather than using an iterative approach native to digital computational architectures.

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105. Linear Coordinate-Descent Message Passing for Quadratic Optimization

Guoqiang Zhang, Richard Heusdens

https://doi.org/10.1162/neco_a_00368 | pp. 3340-3370

In this letter, we propose a new message-passing algorithm for quadratic optimization. The design of the new algorithm is based on linear coordinate descent between neighboring nodes. The updating messages are in a form of linear functions as compared to the min-sum algorithm of which the messages are in a form of quadratic functions. As a result, the linear coordinate-descent (LiCD) algorithm transmits only one parameter per message as opposed to the min-sum algorithm, which transmits two parameters per message. We show that when the quadratic matrix is walk-summable, the LiCD algorithm converges. By taking the LiCD algorithm as a subroutine, we also fix the convergence issue for a general quadratic matrix. The LiCD algorithm works in either a synchronous or asynchronous message-passing manner. Experimental results show that for a general graph with multiple cycles, the LiCD algorithm has comparable

convergence speed to the min-sum algorithm, thereby reducing the number of parameters to be transmitted and the computational complexity.

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106. Active Subspace: Toward Scalable Low-Rank Learning

Guangcan Liu, Shuicheng Yan

https://doi.org/10.1162/neco_a_00369 | pp. 3371-3394

We address the scalability issues in low-rank matrix learning problems. Usually these problems resort to solving nuclear norm regularized optimization problems (NNROPs), which often suffer from high computational complexities if based on existing solvers, especially in large-scale settings. Based on the fact that the optimal solution matrix to an NNROP is often low rank, we revisit the classic mechanism of low-rank matrix factorization, based on which we present an active subspace algorithm for efficiently solving NNROPs by transforming large-scale NNROPs into small-scale problems. The transformation is achieved by factorizing the large solution matrix into the product of a small orthonormal matrix (active subspace) and another small matrix. Although such a transformation generally leads to nonconvex problems, we show that a suboptimal solution can be found by the augmented Lagrange alternating direction method. For the robust PCA (RPCA) (Candès, Li, Ma, & Wright, 2009) problem, a typical example of NNROPs, theoretical results verify the suboptimality of the solution produced by our algorithm. For the general NNROPs, we empirically show that our algorithm significantly reduces the computational complexity without loss of optimality.